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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Alternating Current

JEE Main Physics · Previous Year Questions with Answer Key

204	151	2002 – 2026	9	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

MCQ (Single Correct Answer)	138 questions
Numerical	57 questions
Out of syllabus	9 questions
Answer key	at the end

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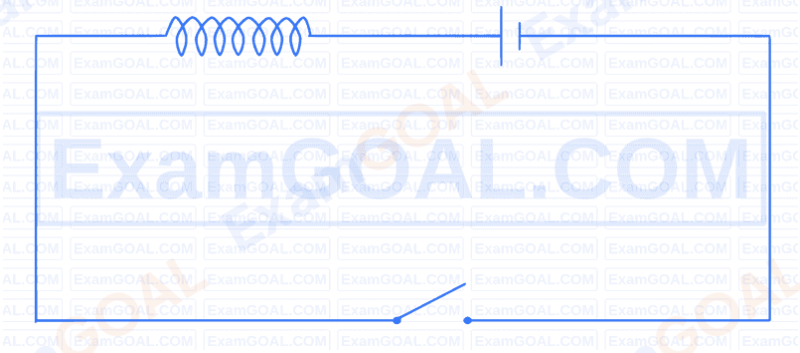
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MCQ (Single Correct Answer) 138 questions

Q1. JEE Main 2026 (Online) 6th April Morning Shift

An inductor of inductance 10 mH having resistance of 100Ω is connected to battery of E.M.F. 1.0 V through a switch as shown in the figure below. After switch is closed, the ratio of instantaneous voltages across the inductor when the current passing through it is 2 mA and 4 mA is ____ .

Open solution



(A) 4/3 (B) 3/4 (C) 5/3 (D) 3/5

Q2. JEE Main 2026 (Online) 6th April Morning Shift

A LCR series circuit driven with $E_{rms} = 90\ \text{V}$ at frequency $f_{\text{d}} = 30\ \text{Hz}$ has resistance $R = 80\Omega$, an inductance with inductive reactance $X_L = 20.0\Omega$ and capacitance with capacitive reactance $X_C = 80.0\Omega$. The power factor of the circuit is ____ .

Open solution

(A) 0.8 (B) 0.64 (C) 0.9 (D) 0.5

Q3. JEE Main 2026 (Online) 5th April Morning Shift

An a.c. source of angular frequency ω is connected across a resistor R and a capacitor C in series. The current is observed as I. Now the frequency of the source is changed to $\omega/4$, (keeping the voltage unchanged) the current is found to be $I/3$. The ratio of resistance to reactance at frequency ω is

Open solution

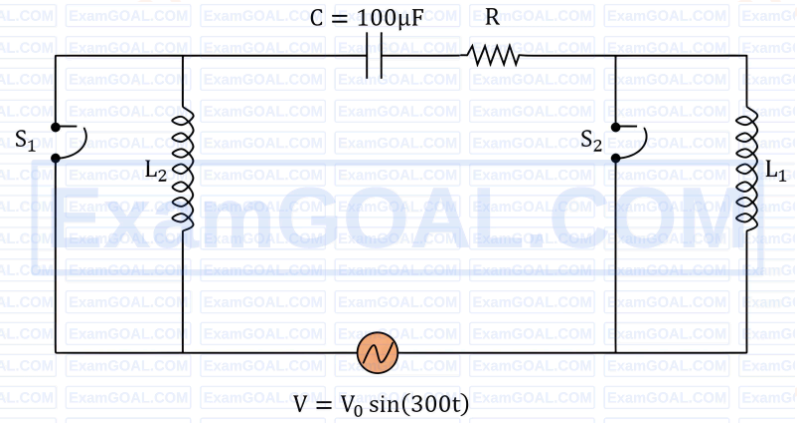
(A) $\sqrt{\frac{6}{7}}$ (B) $\sqrt{\frac{3}{5}}$ (C) $\sqrt{\frac{7}{8}}$ (D) $\sqrt{\frac{3}{4}}$

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Q4. JEE Main 2026 (Online) 2nd April Evening Shift

The figure given below shows an *LCR* series circuit with two switches S_1 and S_2. When switch S_1 is closed keeping S_2 open, the phase difference (φ) between the current and source voltage is 30° and phase difference is 60° when S_2 is closed keeping S_1 open. The value of $(3L_1 - L_2)$ is _____ H.

Open solution



(A) $\frac{9}{2}$ (B) $\frac{2}{9}$ (C) $\frac{1}{3}$ (D) 3

Q5. JEE Main 2026 (Online) 28th January Morning Shift

The electric current in the circuit is given as $i = i_o(t/T)$. The r.m.s current for the period $t = 0$ to $t = T$ is ____ .

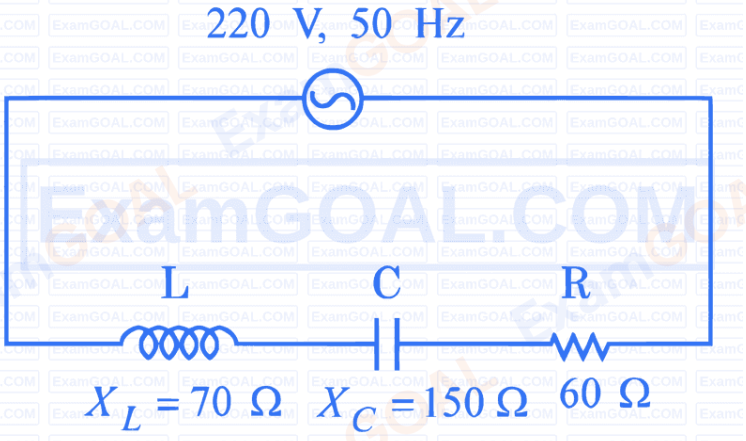
Open solution

(A) $\frac{i_o}{\sqrt{2}}$ (B) $\frac{i_o}{\sqrt{3}}$ (C) i_o (D) $\frac{i_o}{\sqrt{6}}$

Q6. JEE Main 2026 (Online) 24th January Morning Shift

For the series LCR circuit connected with $220\ \text{V}, 50\ \text{Hz}$ a.c source as shown in the figure, the power factor is $\frac{\alpha}{10}$. The value of α is ____ .

Open solution

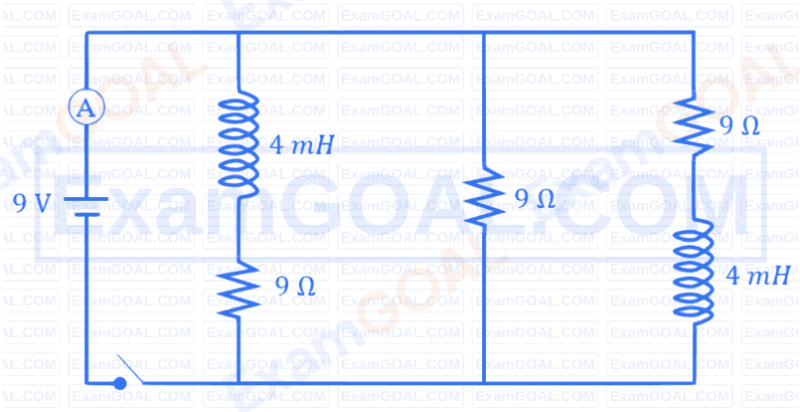


(A) 4 (B) 8 (C) 6 (D) 10

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Q7. JEE Main 2026 (Online) 22nd January Evening Shift

Figure shows the circuit that contains three resistances (9Ω each) and two inductors (4 mH each). The reading of ammeter at the moment switch K is turned ON , is ____ A.



Open solution

(A) 2 (B) 3 (C) 1 (D) zero

Q8. JEE Main 2026 (Online) 21st January Evening Shift

A capacitor C is first charged fully with potential difference of V_0 and disconnected from the battery. The charged capacitor is connected across an inductor having inductance L. In t s, 25% of the initial energy in the capacitor is transferred to the inductor. The value of t is _____ s.

Open solution

(A) $\frac{\pi\sqrt{LC}}{3}$ (B) $\frac{\pi\sqrt{LC}}{2}$ (C) $\pi\sqrt{\frac{LC}{2}}$ (D) $\frac{\pi\sqrt{LC}}{6}$

Q9. JEE Main 2025 (Online) 7th April Morning Shift

An ac current is represented as

$i = 5\sqrt{2} + 10\cos\left(650\pi t + \frac{\pi}{6}\right) Amp$

The r.m.s value of the current is

Open solution

(A) 10 Amp (B) $5\sqrt{2}$ Amp (C) 100 Amp (D) 50 Amp

Q10. JEE Main 2025 (Online) 4th April Morning Shift

An alternating current is represented by the equation, $i = 100\sqrt{2}\sin(100\pi t)$ ampere. The RMS value of current and the frequency of the given alternating current are

Open solution

(A) $\frac{100}{\sqrt{2}}$ A, 100 Hz (B) $50\sqrt{2}$ A, 50 Hz

(C) $100\sqrt{2}$ A, 100 Hz (D) 100 A, 50 Hz

Q11. JEE Main 2025 (Online) 3rd April Evening Shift

An electric bulb rated as 100 W – 220 V is connected to an ac source of rms voltage 220 V. The peak value of current through the bulb is :

Open solution

(A) 0.32 A (B) 0.64 A (C) 0.45 A (D) 2.2 A

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Q12. JEE Main 2025 (Online) 29th January Morning Shift

Open solution

Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Choke coil is simply a coil having a large inductance but a small resistance. Choke coils are used with fluorescent mercury-tube fittings. If household electric power is directly connected to a mercury tube, the tube will be damaged.

Reason (R): By using the choke coil, the voltage across the tube is reduced by a factor $\left(R/\sqrt{R^2+\omega^2L^2}\right)$, where ω is frequency of the supply across resistor R and inductor L. If the choke coil were not used, the voltage across the resistor would be the same as the applied voltage.

In the light of the above statements, choose the **most appropriate answer** from the options given below :

(A) (A) is true but (R) is false

(B) Both $(\mathbf{A})$ and $(\mathbf{R})$ are true and $(\mathbf{R})$ is the correct explanation of $(\mathbf{A})$ 3.

(C) Both $(\mathbf{A})$ and $(\mathbf{R})$ are true but $(\mathbf{R})$ is not the correct explanation of $(\mathbf{A})$

(D) (A) is false but (R) is true

Q13. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

An alternating current is given by $\mathrm{I} = \mathrm{I_A}\sin\omega\mathrm{t} + \mathrm{I_B}\cos\omega\mathrm{t}$. The r.m.s current will be

(A) $\frac{\sqrt{\mathrm{I_A^2}+\mathrm{I_B^2}}}{2}$

(B) $\sqrt{\frac{\mathrm{I_A^2}+\mathrm{I_B^2}}{2}}$

(C) $\frac{|\mathrm{I_A}+\mathrm{I_B}|}{\sqrt{2}}$

(D) $\sqrt{\mathrm{I_A^2}+\mathrm{I_B^2}}$

Q14. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

A series LCR circuit is connected to an alternating source of emf E. The current amplitude at resonant frequency is I_0. If the value of resistance R becomes twice of its initial value then amplitude of current at resonance will be

(A) $\frac{\mathrm{I_0}}{2}$ **(B)** $\frac{\mathrm{I_0}}{\sqrt{2}}$ **(C)** $2\mathrm{I_0}$ **(D)** $\mathrm{I_0}$

Q15. JEE Main 2024 (Online) 9th April Morning Shift

Open solution

A bulb and a capacitor are connected in series across an ac supply. A dielectric is then placed between the plates of the capacitor. The glow of the bulb :

(A) becomes zero **(B)** remains same **(C)** increases **(D)** decreases

Q16. JEE Main 2024 (Online) 8th April Evening Shift

Open solution

A coil of negligible resistance is connected in series with 90Ω resistor across $120\ \mathrm{V}, 60\ \mathrm{Hz}$ supply. A voltmeter reads $36\ \mathrm{V}$ across resistance. Inductance of the coil is :

(A) 0.91 H **(B)** 0.76 H **(C)** 2.86 H **(D)** 0.286 H

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Q17. JEE Main 2024 (Online) 8th April Morning Shift

A LCR circuit is at resonance for a capacitor C, inductance L and resistance R. Now the value of resistance is halved keeping all other parameters same. The current amplitude at resonance will be now:

Open solution

(A) halved (B) same (C) Zero (D) double

Q18. JEE Main 2024 (Online) 6th April Morning Shift

Given below are two statements :

Statement I : In an LCR series circuit, current is maximum at resonance.

Statement II : Current in a purely resistive circuit can never be less than that in a series LCR circuit when connected to same voltage source.

In the light of the above statements, choose the correct answer from the options given below :

Open solution

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q19. JEE Main 2024 (Online) 5th April Evening Shift

A series LCR circuit is subjected to an ac signal of $200\ \mathrm{V}, 50\ \mathrm{Hz}$. If the voltage across the inductor $(\mathrm{L} = 10\ \mathrm{mH})$ is $31.4\ \mathrm{V}$, then the current in this circuit is _____.

Open solution

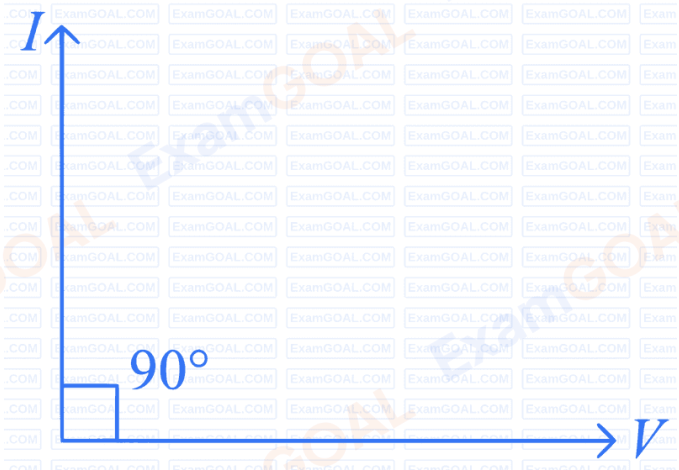
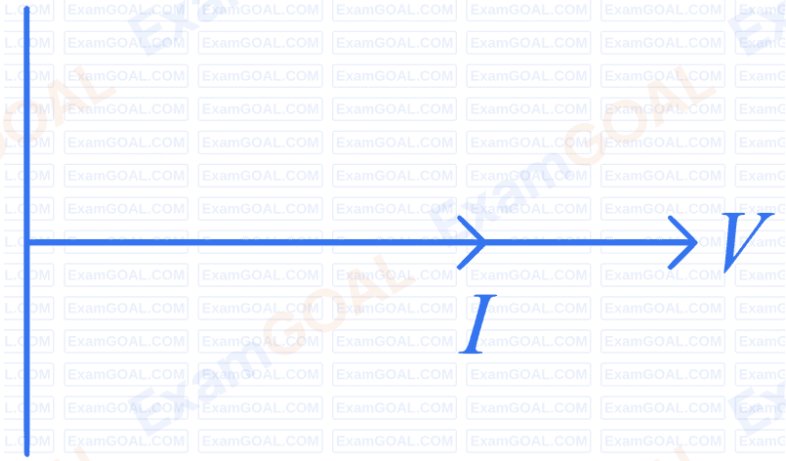
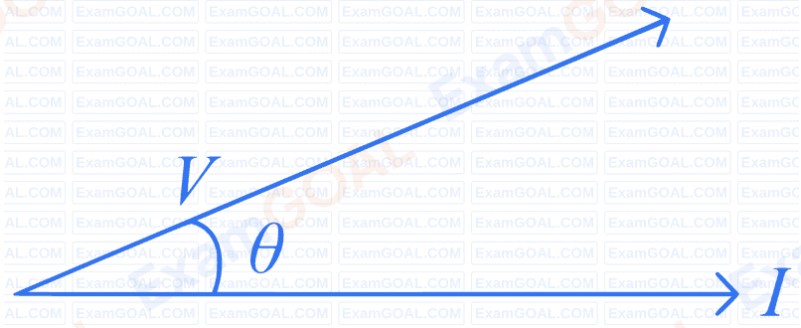
(A) 10 A (B) 10 mA (C) 68 A (D) 63 A

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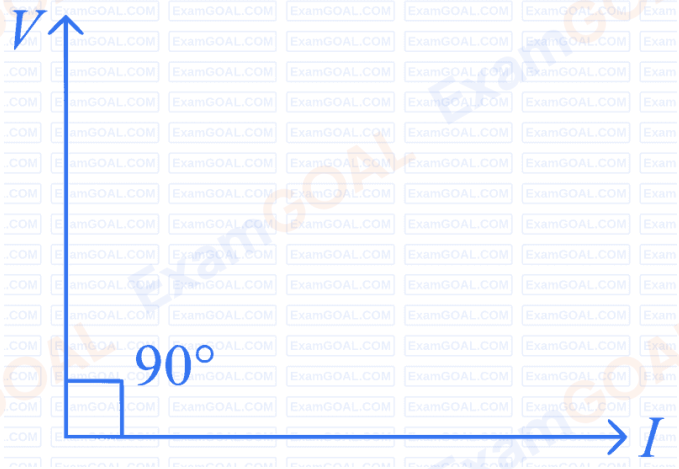
Q20. JEE Main 2024 (Online) 4th April Evening Shift

Open solution

Match List I with List II

	LIST I		LIST II
A.	Purely capacitive circuit	I.	
B.	Purely inductive circuit	II.	
C.	LCR series at resonance	III.	

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	LIST I		LIST II
D.	LCR series circuit	IV.	

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-III, D-II

(B) A-I, B-IV, C-II, D-III

(C) A-IV, B-I, C-II, D-III

(D) A-I, B-IV, C-III, D-II

Q21. JEE Main 2024 (Online) 4th April Morning Shift

In an ac circuit, the instantaneous current is zero, when the instantaneous voltage is maximum. In this case, the source may be connected to :

A. pure inductor.

B. pure capacitor.

C. pure resistor.

D. combination of an inductor and capacitor.

Choose the correct answer from the options given below :

(A) A, B and C only

(B) A and B only

(C) A, B and D only

(D) B, C and D only

Open solution

Q22. JEE Main 2024 (Online) 1st February Evening Shift

A transformer has an efficiency of 80% and works at $10\ \mathrm{V}$ and $4\ \mathrm{kW}$. If the secondary voltage is $240\ \mathrm{V}$, then the current in the secondary coil is :

(A) $1.33\ \mathrm{A}$

(B) $13.33\ \mathrm{A}$

(C) $1.59\ \mathrm{A}$

(D) $15.1\ \mathrm{A}$

Open solution

Q23. JEE Main 2024 (Online) 1st February Morning Shift

In series LCR circuit, the capacitance is changed from C to $4C$. To keep the resonance frequency unchanged, the new inductance should be:

(A) increased by $2\ \mathrm{L}$

(B) reduced by $\frac{1}{4}\ \mathrm{L}$

(C) reduced by $\frac{3}{4}\ \mathrm{L}$

(D) increased to $4\ \mathrm{L}$

Open solution

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Q24. JEE Main 2024 (Online) 31st January Evening Shift

An AC voltage $V = 20\sin 200\pi t$ is applied to a series LCR circuit which drives a current $I = 10\sin\left(200\pi t + \frac{\pi}{3}\right)$. The average power dissipated is:

(A) 21.6 W (B) 200 W (C) 173.2 W (D) 50 W

Open solution

Q25. JEE Main 2024 (Online) 30th January Evening Shift

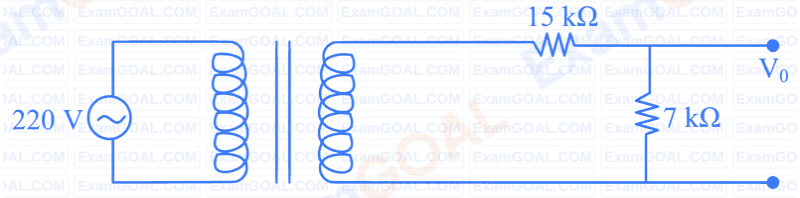
An alternating voltage $V(t) = 220\sin 100\pi t$ volt is applied to a purely resistive load of 50Ω. The time taken for the current to rise from half of the peak value to the peak value is:

(A) 7.2 ms (B) 3.3 ms (C) 5 ms (D) 2.2 ms

Open solution

Q26. JEE Main 2024 (Online) 30th January Morning Shift

Primary coil of a transformer is connected to $220\ \mathrm{V}$ ac. Primary and secondary turns of the transformer are 100 and 10 respectively. Secondary coil of transformer is connected to two series resistances shown in figure. The output voltage (V_0) is :



(A) 7 V (B) 44 V (C) 22 V (D) 15 V

Open solution

Q27. JEE Main 2024 (Online) 30th January Morning Shift

A series L.R circuit connected with an ac source $E = (25\sin 1000t)V$ has a power factor of $\frac{1}{\sqrt{2}}$. If the source of emf is changed to $\mathrm{E} = (20\sin 2000\mathrm{t})\mathrm{V}$, the new power factor of the circuit will be :

(A) $\frac{1}{\sqrt{3}}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{\sqrt{5}}$ (D) $\frac{1}{\sqrt{7}}$

Open solution

Q28. JEE Main 2024 (Online) 29th January Evening Shift

In an a.c. circuit, voltage and current are given by:

$V = 100\sin(100t)V$ and $I = 100\sin\left(100t + \frac{\pi}{3}\right)\mathrm{mA}$ respectively.

The average power dissipated in one cycle is:

(A) 5 W (B) 25 W (C) 2.5 W (D) 10 W

Open solution

Q29. JEE Main 2024 (Online) 29th January Morning Shift

A capacitor of capacitance $100\mu\mathrm{F}$ is charged to a potential of $12\ \mathrm{V}$ and connected to a $6.4\ \mathrm{mH}$ inductor to produce oscillations. The maximum current in the circuit would be :

(A) 2.0 A (B) 3.2 A (C) 1.5 A (D) 1.2 A

Open solution

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Q30. JEE Main 2024 (Online) 27th January Evening Shift

Open solution

Primary side of a transformer is connected to $230\ \mathrm{V}, 50\ \mathrm{Hz}$ supply. Turns ratio of primary to secondary winding is $10:1$. Load resistance connected to secondary side is 46Ω. The power consumed in it is :

(A) 11.5 W (B) 12.5 W (C) 10.0 W (D) 12.0 W

Q31. JEE Main 2023 (Online) 13th April Evening Shift

Open solution

Given below are two statements:

Statement I : An AC circuit undergoes electrical resonance if it contains either a capacitor or an inductor.

Statement II : An AC circuit containing a pure capacitor or a pure inductor consumes high power due to its non-zero power factor.

In the light of above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are false

(B) Statement I is true but statement II is false

(C) Statement I is false but statement II is true

(D) Both Statement I and Statement II are true

Q32. JEE Main 2023 (Online) 12th April Morning Shift

Open solution

Given below are two statements:

Statement I : When the frequency of an a.c source in a series LCR circuit increases, the current in the circuit first increases, attains a maximum value and then decreases.

Statement II : In a series LCR circuit, the value of power factor at resonance is one.

In the light of given statements, choose the most appropriate answer from the options given below.

(A) Both Statement I and Statement II are False.

(B) Statement I is incorrect but Statement II is true.

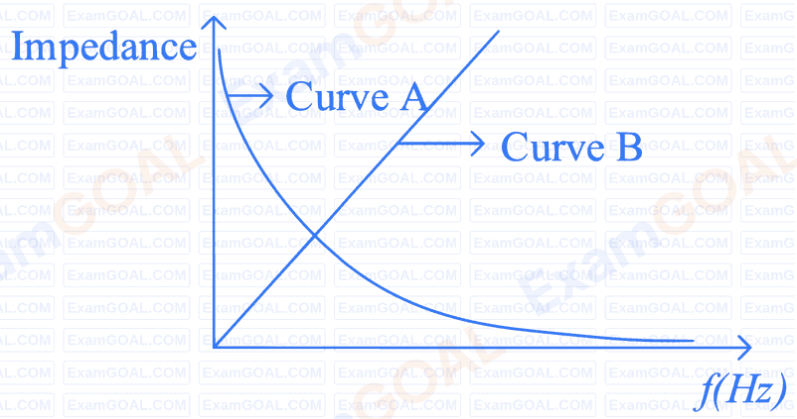
(C) Both Statement I and Statement II are true.

(D) Statement I is correct but Statement II is false.

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Q33. JEE Main 2023 (Online) 11th April Morning Shift

Open solution



As per the given graph, choose the correct representation for curve A and curve B.

Where X_C = reactance of pure capacitive circuit connected with A.C. source

X_L = reactance of pure inductive circuit connected with A.C. source

R = impedance of pure resistive circuit connected with A.C. source.

Z = Impedance of the LCR series circuit

(A) $A = X_L, B = R$

(B) $A = X_L, B = Z$

(C) $A = X_C, B = X_L$

(D) $A = X_C, B = R$

Q34. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

Given below are two statements:

Statement I : Maximum power is dissipated in a circuit containing an inductor, a capacitor and a resistor connected in series with an AC source, when resonance occurs

Statement II : Maximum power is dissipated in a circuit containing pure resistor due to zero phase difference between current and voltage.

In the light of the above statements, choose the correct answer from the options given below:

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are true

Q35. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

A capacitor of capacitance $150.0\ \mu F$ is connected to an alternating source of emf given by $E = 36\sin(120\pi t)V$. The maximum value of current in the circuit is approximately equal to :

(A) $\frac{1}{\sqrt{2}}A$

(B) $2\sqrt{2}A$

(C) $\sqrt{2}A$

(D) $2A$

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Q36. JEE Main 2023 (Online) 1st February Morning Shift

Open solution

Match List - I with List - II :

	List I		List II
A.	AC generator	I.	Presence of both L and C
B.	Transformer	II.	Electromagnetic Induction
C.	Resonance phenomenon to occur	III.	Quality factor
D.	Sharpness of resonance	IV.	Mutual Induction

Choose the correct answer from the options given below :

(A) A-II, B-I, C-III, D-IV

(B) A-IV, B-III, C-I, D-II

(C) A-II, B-IV, C-I, D-III

(D) A-IV, B-II, C-I, D-III

Q37. JEE Main 2023 (Online) 31st January Evening Shift

Open solution

An alternating voltage source $V = 260\sin(628t)$ is connected across a pure inductor of 5mH. Inductive reactance in the circuit is :

(A) 6.28Ω

(B) 0.318Ω

(C) 0.5Ω

(D) 3.14Ω

Q38. JEE Main 2023 (Online) 31st January Morning Shift

Open solution

If $\mathrm{R}, \mathrm{X_L}$, and $\mathrm{X_C}$ represent resistance, inductive reactance and capacitive reactance. Then which of the following is dimensionless :

(A) $\frac{R}{X_L X_C}$

(B) RX_LX_C

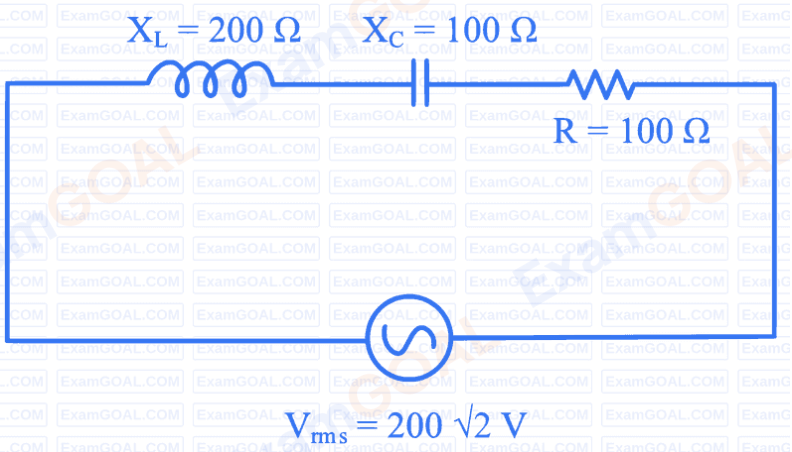
(C) $\frac{R}{\sqrt{X_L X_C}}$

(D) $R\frac{X_L}{X_C}$

Q39. JEE Main 2023 (Online) 30th January Evening Shift

Open solution

In the given circuit, rms value of current (I_{rms}) through the resistor R is:



(A) $2\sqrt{2}\,\text{A}$

(B) $2\,\text{A}$

(C) $\frac{1}{2}\,\text{A}$

(D) $20\,\text{A}$

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Q40. JEE Main 2023 (Online) 30th January Morning Shift

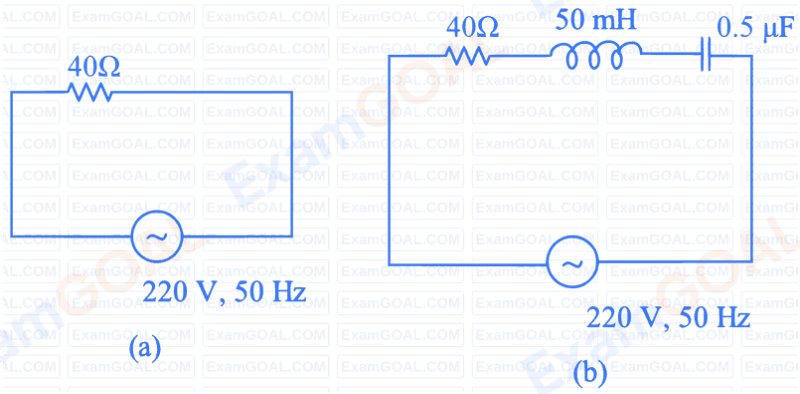
In a series LR circuit with $X_L = R$, power factor P_1. If a capacitor of capacitance C with $X_C = X_L$ is added to the circuit the power factor becomes P_2. The ratio of P_1 to P_2 will be :

(A) $1 : \sqrt{2}$ (B) 1 : 3 (C) 1 : 2 (D) 1 : 1

Open solution

Q41. JEE Main 2023 (Online) 29th January Evening Shift

For the given figures, choose the correct options :



(A) The rms current in circuit (b) can be larger than that in (a)

(B) The rms current in figure (a) is always equal to that in figure (b)

(C) The rms current in circuit (b) can never be larger than that in (a)

(D) At resonance, current in (b) is less than that in (a)

Open solution

Q42. JEE Main 2023 (Online) 25th January Morning Shift

In an LC oscillator, if values of inductance and capacitance become twice and eight times, respectively, then the resonant frequency of oscillator becomes x times its initial resonant frequency ω_0. The value of x is :

(A) 1/4 (B) 1/16 (C) 4 (D) 16

Open solution

Q43. JEE Main 2022 (Online) 30th July Evening Shift

In an A.C circuit, V & I are given by $V = 250\sin(100t)$ volt and $I = 10\sin\left(100t + \frac{\pi}{3}\right)$ A respectively

The power dissipated in the circuit is:

(A) 2500 W (B) 625 W (C) 1250 W (D) $625\sqrt{2}$ W

Open solution

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Q44. JEE Main 2022 (Online) 29th July Evening Shift

A circuit element X when connected to an a.c. supply of peak voltage $100\ \mathrm{V}$ gives a peak current of $5\ \mathrm{A}$ which is in phase with the voltage. A second element Y when connected to the same a.c. supply also gives the same value of peak current which lags behind the voltage by $\frac{\pi}{2}$. If X and Y are connected in series to the same supply, what will be the rms value of the current in ampere?

Open solution

(A) $\frac{10}{\sqrt{2}}$ (B) $\frac{5}{\sqrt{2}}$ (C) $5\sqrt{2}$ (D) $\frac{5}{2}$

Q45. JEE Main 2022 (Online) 29th July Morning Shift

An alternating emf $\mathrm{E} = 440\sin 100\pi\mathrm{t}$ is applied to a circuit containing an inductance of $\frac{\sqrt{2}}{\pi}\mathrm{H}$. If an a.c. ammeter is connected in the circuit, its reading will be :

Open solution

(A) 4.4 A (B) 1.55 A (C) 2.2 A (D) 3.11 A

Q46. JEE Main 2022 (Online) 29th July Morning Shift

A coil of inductance 1 H and resistance $100\,\Omega$ is connected to a battery of 6 V. Determine approximately :

Open solution

(a) The time elapsed before the current acquires half of its steady - state value.

(b) The energy stored in the magnetic field associated with the coil at an instant 15 ms after the circuit is switched on. (Given $\ln 2 = 0.693, \mathrm{e}^{-3/2} = 0.25$)

(A) t = 10 ms; U = 2 mJ (B) t = 10 ms; U = 1 mJ

(C) t = 7 ms; U = 1 mJ (D) t = 7 ms; U = 2 mJ

Q47. JEE Main 2022 (Online) 28th July Evening Shift

A transformer operating at primary voltage $8\,\mathrm{kV}$ and secondary voltage $160\ \mathrm{V}$ serves a load of $80\,\mathrm{kW}$. Assuming the transformer to be ideal with purely resistive load and working on unity power factor, the loads in the primary and secondary circuit would be

Open solution

(A) $800\,\Omega$ and $1.06\,\Omega$ (B) $10\,\Omega$ and $500\,\Omega$

(C) $800\,\Omega$ and $0.32\,\Omega$ (D) $1.06\,\Omega$ and $500\,\Omega$

Q48. JEE Main 2022 (Online) 28th July Morning Shift

The equation of current in a purely inductive circuit is $5\sin(49\,\pi t - 30^\circ)$. If the inductance is $30\,\mathrm{mH}$ then the equation for the voltage across the inductor, will be :

Open solution

$\{$ Let $\pi = \frac{22}{7}\}$

(A) $1.47\sin(49\pi t - 30^\circ)$ (B) $1.47\sin(49\pi t + 60^\circ)$

(C) $23.1\sin(49\pi t - 30^\circ)$ (D) $23.1\sin(49\pi t + 60^\circ)$

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Q49. JEE Main 2022 (Online) 27th July Evening Shift

A series LCR circuit has $\mathrm{L} = 0.01\,\mathrm{H}, \mathrm{R} = 10\,\Omega$ and $\mathrm{C} = 1\mu\mathrm{F}$ and it is connected to ac voltage of amplitude $(\mathrm{V_m})50\ \mathrm{V}$. At frequency 60% lower than resonant frequency, the amplitude of current will be approximately :

(A) 466 mA (B) 312 mA (C) 238 mA (D) 196 mA

Open solution

Q50. JEE Main 2022 (Online) 27th July Morning Shift

A direct current of $4\ \mathrm{A}$ and an alternating current of peak value $4\ \mathrm{A}$ flow through resistance of $3\,\Omega$ and $2\,\Omega$ respectively. The ratio of heat produced in the two resistances in same interval of time will be :

(A) 3 : 2 (B) 3 : 1 (C) 3 : 4 (D) 4 : 3

Open solution

Q51. JEE Main 2022 (Online) 26th July Morning Shift

In a series LR circuit $X_L = R$ and power factor of the circuit is P_1. When capacitor with capacitance C such that $X_L = X_C$ is put in series, the power factor becomes P_2. The ratio $\frac{P_1}{P_2}$ is:

(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{\sqrt{3}}{\sqrt{2}}$ (D) 2 : 1

Open solution

Q52. JEE Main 2022 (Online) 25th July Evening Shift

When you walk through a metal detector carrying a metal object in your pocket, it raises an alarm. This phenomenon works on :

(A) Electromagnetic induction (B) Resonance in ac circuits

(C) Mutual induction in ac circuits (D) Interference of electromagnetic waves

Open solution

Q53. JEE Main 2022 (Online) 25th July Morning Shift

To increase the resonant frequency in series LCR circuit,

(A) source frequency should be increased.

(B) another resistance should be added in series with the first resistance.

(C) another capacitor should be added in series with the first capacitor.

(D) the source frequency should be decreased.

Open solution

Q54. JEE Main 2022 (Online) 25th July Morning Shift

The rms value of conduction current in a parallel plate capacitor is $6.9\,\mu\mathrm{A}$. The capacity of this capacitor, if it is connected to $230\ \mathrm{V}$ ac supply with an angular frequency of $600\,\mathrm{rad/s}$, will be :

(A) 5 pF (B) 50 pF (C) 100 pF (D) 200 pF

Open solution

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Q55. JEE Main 2022 (Online) 29th June Morning Shift

Open solution

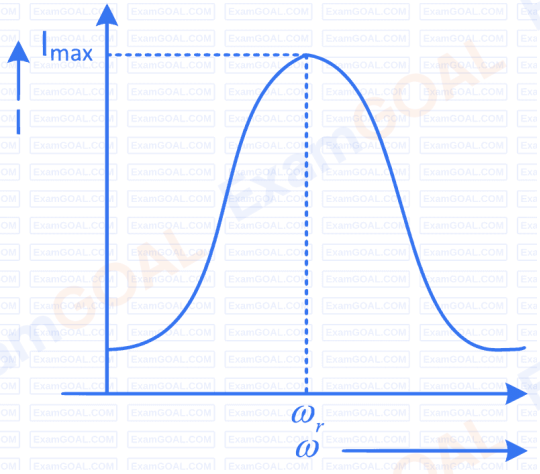
For a series LCR circuit, I vs ω curve is shown :

(a) To the left of ω_r, the circuit is mainly capacitive.

(b) To the left of ω_r, the circuit is mainly inductive.

(c) At ω_r, impedance of the circuit is equal to the resistance of the circuit.

(d) At ω_r, impedance of the circuit is 0.



Choose the most appropriate answer from the options given below :

(A) (a) and (d) only.

(B) (b) and (d) only.

(C) (a) and (c) only.

(D) (b) and (c) only.

Q56. JEE Main 2022 (Online) 27th June Evening Shift

Open solution

If L, C and R are the self inductance, capacitance and resistance respectively, which of the following does not have the dimension of time?

(A) RC

(B) $\frac{L}{R}$

(C) $\sqrt{LC}$

(D) $\frac{L}{C}$

Q57. JEE Main 2022 (Online) 27th June Morning Shift

Open solution

The current flowing through an ac circuit is given by

I = 5 sin(120πt)A

How long will the current take to reach the peak value starting from zero?

(A) $\frac{1}{60}$ s

(B) 60 s

(C) $\frac{1}{120}$ s

(D) $\frac{1}{240}$ s

Q58. JEE Main 2022 (Online) 25th June Evening Shift

Open solution

A sinusoidal voltage V(t) = 210 sin 3000 t volt is applied to a series LCR circuit in which L = 10 mH, C = 25 μF and R = 100 Ω. The phase difference (Φ) between the applied voltage and resultant current will be :

(A) $\tan^{-1}(0.17)$

(B) $\tan^{-1}(9.46)$

(C) $\tan^{-1}(0.30)$

(D) $\tan^{-1}(13.33)$

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Q59. JEE Main 2022 (Online) 25th June Morning Shift

Open solution

Match List-I with List-II.

	List - I		List -II
(A)	AC generator	(I)	Detects the presence of current in the circuit
(B)	Galvanometer	(II)	Converts mechanical energy into electrical energy
(C)	Transformer	(III)	Works on the principle of resonance in AC circuit
(D)	Metal detector	(IV)	Changes an alternating voltage for smaller or greater value

Choose the correct answer from the options given below :

(A) (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

(B) (A) - (II), (B) - (I), (C) - (III), (D) - (IV)

(C) (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

(D) (A) - (III), (B) - (I), (C) - (II), (D) - (IV)

Q60. JEE Main 2022 (Online) 25th June Morning Shift

Open solution

If wattless current flows in the AC circuit, then the circuit is :

(A) Purely Resistive circuit

(B) Purely Inductive circuit

(C) LCR series circuit

(D) RC series circuit only

Q61. JEE Main 2022 (Online) 24th June Evening Shift

Open solution

Given below are two statements :

Statement I : The reactance of an ac circuit is zero. It is possible that the circuit contains a capacitor and an inductor.

Statement II : In ac circuit, the average power delivered by the source never becomes zero.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

Q62. JEE Main 2022 (Online) 24th June Morning Shift

Open solution

A resistance of 40 Ω is connected to a source of alternating current rated 220 V, 50 Hz. Find the time taken by the current to change from its maximum value to the rms value :

(A) 2.5 ms

(B) 1.25 ms

(C) 2.5 s

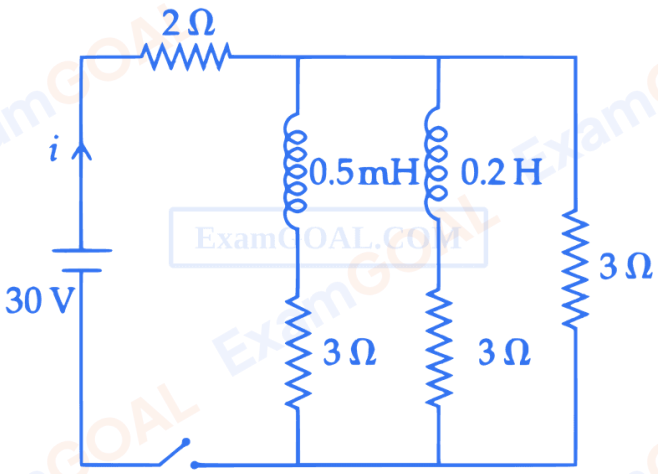
(D) 0.25 s

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Q63. JEE Main 2021 (Online) 1st September Evening Shift

For the given circuit the current i through the battery when the key is closed and the steady state has been reached is _____.

Open solution



(A) 6 A (B) 25 A (C) 10 A (D) 0 A

Q64. JEE Main 2021 (Online) 31st August Morning Shift

In an ac circuit, an inductor, a capacitor and a resistor are connected in series with $X_L = R = X_C$. Impedance of this circuit is :

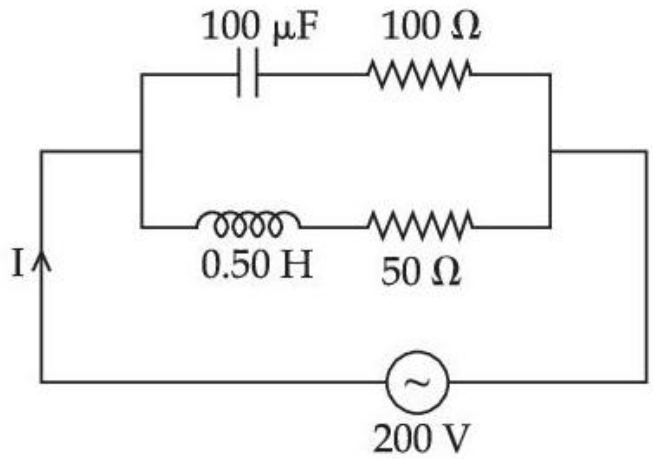
Open solution

(A) $2R^2$ (B) Zero (C) R (D) $R\sqrt{2}$

Q65. JEE Main 2021 (Online) 26th August Evening Shift

In the given circuit the AC source has $\omega = 100$ rad s^{-1}. Considering the inductor and capacitor to be ideal, what will be the current I flowing through the circuit?

Open solution



(A) 5.9 A (B) 3.16 A (C) 0.94 A (D) 6 A

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Q66. JEE Main 2021 (Online) 26th August Morning Shift

A series LCR circuit driven by 300 V at a frequency of 50 Hz contains a resistance R = 3 kΩ, an inductor of inductive reactance X_L = 250 $\pi\Omega$ and an unknown capacitor. The value of capacitance to maximize the average power should be : (Take π^2 = 10)

Open solution

(A) 4 μF (B) 25 μF (C) 400 μF (D) 40 μF

Q67. JEE Main 2021 (Online) 27th July Evening Shift

A 100Ω resistance, a 0.1 μF capacitor and an inductor are connected in series across a 250 V supply at variable frequency. Calculate the value of inductance of inductor at which resonance will occur. Given that the resonant frequency is 60 Hz.

Open solution

(A) 0.70 H (B) 70.3 mH (C) 7.03×10^{-5} H (D) 70.3 H

Q68. JEE Main 2021 (Online) 27th July Morning Shift

A 0.07 H inductor and a 12Ω resistor are connected in series to a 220V, 50 Hz ac source. The approximate current in the circuit and the phase angle between current and source voltage are respectively. [Take π as $\frac{22}{7}$]

Open solution

(A) 8.8 A and $\tan^{-1}\left(\frac{11}{6}\right)$ (B) 88 A and $\tan^{-1}\left(\frac{11}{6}\right)$

(C) 0.88 A and $\tan^{-1}\left(\frac{11}{6}\right)$ (D) 8.8 A and $\tan^{-1}\left(\frac{6}{11}\right)$

Q69. JEE Main 2021 (Online) 25th July Evening Shift

A 10 Ω resistance is connected across 220V $-$ 50 Hz AC supply. The time taken by the current to change from its maximum value to the rms value is :

Open solution

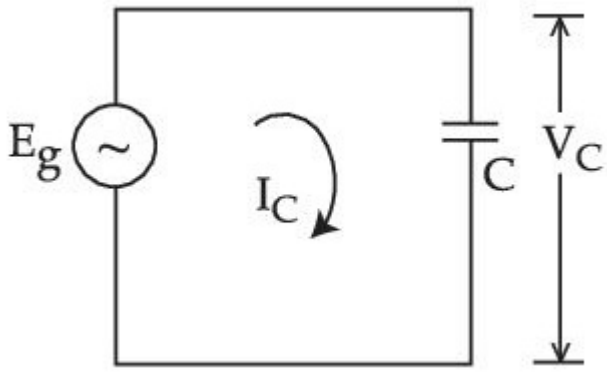
(A) 2.5 ms (B) 1.5 ms (C) 3.0 ms (D) 4.5 ms

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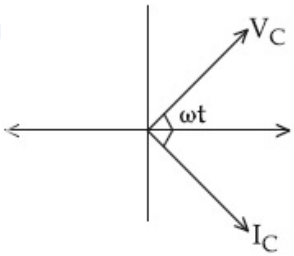
Q70. JEE Main 2021 (Online) 22th July Evening Shift

In a circuit consisting of a capacitance and a generator with alternating emf $E_g = E_{g_0} \sin\omega t$, V_C and I_C are the voltage and current. Correct phasor diagram for such circuit is

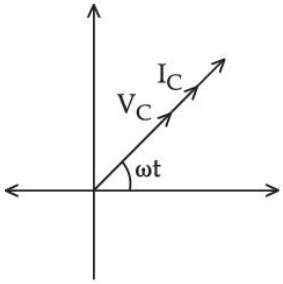
Open solution



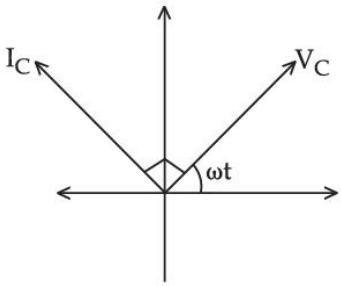
(A)



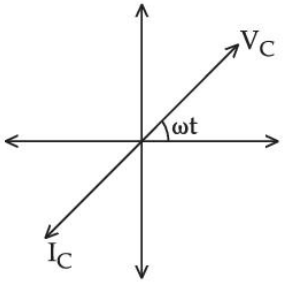
(B)



(C)



(D)



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Q71. JEE Main 2021 (Online) 22th July Evening Shift

Open solution

Match List - I with List - II

	List - I		List - II
(a)	$\omega L > \frac{1}{\omega C}$	(i)	Current is in phase with emf
(b)	$\omega L = \frac{1}{\omega C}$	(ii)	Current lags behind the applied emf
(c)	$\omega L < \frac{1}{\omega C}$	(iii)	Maximum current occurs
(d)	Resonant frequency	(iv)	Current leads the emf

Choose the correct answer from the options given below

(A) a(ii), b(i), c(iv), d(iii)

(B) a(ii), b(i), c(iii), d(iv)

(C) a(iii), b(i), c(iv), d(ii)

(D) a(iv), b(iii), c(ii), d(i)

Q72. JEE Main 2021 (Online) 20th July Evening Shift

Open solution

For a series LCR circuit with R = 100 Ω, L = 0.5 mH and C = 0.1 pF connected across 220V$-$50 Hz AC supply, the phase angle between current and supplied voltage and the nature of the circuit is :

(A) 0°, resistive circuit

(B) $\approx$ 90°, predominantly inductive circuit

(C) 0°, resonance circuit

(D) $\approx$ 90°, predominantly capacitive circuit

Q73. JEE Main 2021 (Online) 18th March Evening Shift

Open solution

In a series LCR circuit, the inductive reactance (X_L) is 10Ω and the capacitive reactance (X_C) is 4Ω. The resistance (R) in the circuit is 6Ω. The power factor of the circuit is :

(A) $\frac{1}{2}$ **(B)** $\frac{\sqrt{3}}{2}$ **(C)** $\frac{1}{\sqrt{2}}$ **(D)** $\frac{1}{2\sqrt{2}}$

Q74. JEE Main 2021 (Online) 18th March Morning Shift

Open solution

In a series LCR resonance circuit, if we change the resistance only, from a lower to higher value :

(A) The bandwidth of resonance circuit will increase.

(B) The resonance frequency will increase.

(C) The quality factor will increase.

(D) The quality factor and the resonance frequency will remain constant.

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Q75. JEE Main 2021 (Online) 18th March Morning Shift

An AC source rated 220 V, 50 Hz is connected to a resistor. The time taken by the current to change from its maximum to the rms value is :

(A) 2.5 ms (B) 25 ms (C) 2.5 s (D) 0.25 ms

Open solution

Q76. JEE Main 2021 (Online) 17th March Evening Shift

Match List - I with List - II

	List - I		List - II
(a)	Phase difference between current and voltage in a purely resistive AC circuit	(i)	$\frac{\pi}{2}$; current leads voltage
(b)	Phase difference between current and voltage in a pure inductive AC circuit	(ii)	zero
(c)	Phase difference between current and voltage in a pure capacitive AC circuit	(iii)	$\frac{\pi}{2}$; current lags voltage
(d)	Phase difference between current and voltage in an LCR series circuit	(iv)	$\tan^{-1}\left(\frac{X_C - X_L}{R}\right)$

Open solution

Choose the most appropriate answer from the options given below :

(A) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)
(B) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)
(C) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
(D) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

Q77. JEE Main 2021 (Online) 17th March Evening Shift

What happens to the inductive reactance and the current in a purely inductive circuit if the frequency is halved?

(A) Both, inducting reactance and current will be doubled.
(B) Inductive reactance will be doubled and current will be halved.
(C) Both, inductive reactance and current will be halved.
(D) Inductive reactance will be halved and current will be doubled.

Open solution

Q78. JEE Main 2021 (Online) 17th March Morning Shift

An AC current is given by $I = I_1 \sin\omega t + I_2 \cos\omega t$. A hot wire ammeter will give a reading :

(A) $\frac{I_1+I_2}{\sqrt{2}}$ (B) $\sqrt{\frac{I_1^2-I_2^2}{2}}$
(C) $\sqrt{\frac{I_1^2+I_2^2}{2}}$ (D) $\frac{I_1+I_2}{2\sqrt{2}}$

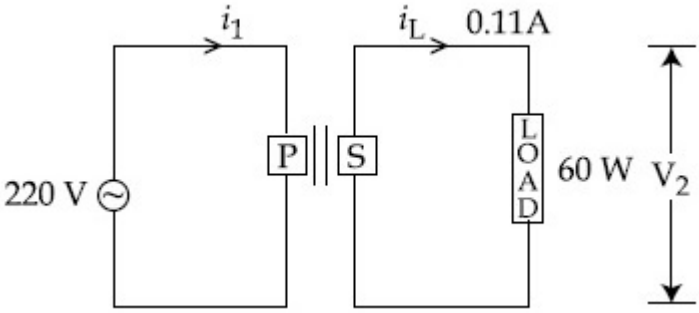
Open solution

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Q79. JEE Main 2021 (Online) 16th March Evening Shift

Open solution

For the given circuit, comment on the type of transformer used.



(A) Auxiliary transformer

(B) Step down transformer

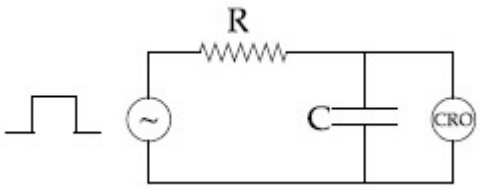
(C) Step-up transformer

(D) Auto transformer

Q80. JEE Main 2021 (Online) 16th March Morning Shift

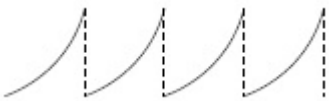
Open solution

An RC circuit as shown in the figure is driven by a AC source generating a square wave. The output wave pattern monitored by CRO would look close to :



(A) 

(B) 

(C) 

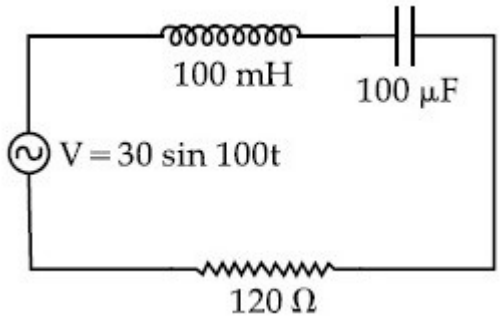
(D) 

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Q81. JEE Main 2021 (Online) 26th February Evening Shift

Find the peak current and resonant frequency of the following circuit (as shown in figure).

Open solution



(A) 2 A and 100 Hz (B) 2 A and 50 Hz (C) 0.2 A and 100 Hz (D) 0.2 A and 50 Hz

Q82. JEE Main 2021 (Online) 26th February Morning Shift

An alternating current is given by the equation i = i_1 sin ωt + i_2 cos ωt. The rms current will be :

Open solution

(A) $\frac{1}{\sqrt{2}}\left(i_1^2+i_2^2\right)^{\frac{1}{2}}$ (B) $\frac{1}{\sqrt{2}}(i_1+i_2)$

(C) $\frac{1}{\sqrt{2}}(i_1+i_2)^2$ (D) $\frac{1}{2}\left(i_1^2+i_2^2\right)^{\frac{1}{2}}$

Q83. JEE Main 2021 (Online) 25th February Evening Shift

An LCR circuit contains resistance of 110Ω and a supply of 220 V at 300 rad/s angular frequency. If only capacitance is removed from the circuit, current lags behind the voltage by 45°. If on the other hand, only inductor is removed the current leads by 45° with the applied voltage. The rms current flowing in the circuit will be :

Open solution

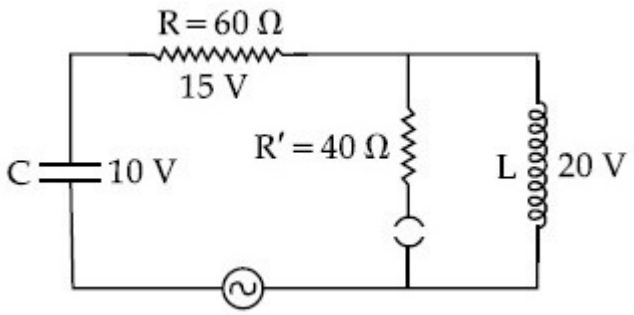
(A) 1 A (B) 2.5 A (C) 2 A (D) 1.5 A

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Q84. JEE Main 2021 (Online) 25th February Morning Shift

Open solution

The angular frequency of alternating current in a L-C-R circuit is 100 rad/s. The components connected are shown in the figure. Find the value of inductance of the coil and capacity of condenser.



(A) 0.8 H and 150 μF

(B) 0.8 H and 250 μF

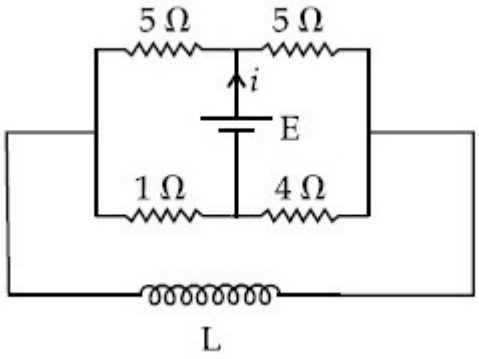
(C) 1.33 H and 150 μF

(D) 1.33 H and 250 μF

Q85. JEE Main 2021 (Online) 25th February Morning Shift

Open solution

The current (i) at time t = 0 and t = ∞ respectively for the given circuit is :



(A) $\frac{18E}{55}, \frac{5E}{18}$

(B) $\frac{5E}{18}, \frac{18E}{55}$

(C) $\frac{5E}{18}, \frac{10E}{33}$

(D) $\frac{10E}{33}, \frac{5E}{18}$

Q86. JEE Main 2020 (Online) 4th September Evening Slot

Open solution

A series L-R circuit is connected to a battery of emf V. If the circuit is switched on at t = 0, then the time at which the energy stored in the inductor reaches $\left(\frac{1}{n}\right)$ times of its maximum value, is :

(A) $\frac{L}{R}\ln\left(\frac{\sqrt{n}}{\sqrt{n}+1}\right)$

(B) $\frac{L}{R}\ln\left(\frac{\sqrt{n}}{\sqrt{n}-1}\right)$

(C) $\frac{L}{R}\ln\left(\frac{\sqrt{n}+1}{\sqrt{n}-1}\right)$

(D) $\frac{L}{R}\ln\left(\frac{\sqrt{n}-1}{\sqrt{n}}\right)$

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Q87. JEE Main 2020 (Online) 3rd September Morning Slot

A 750 Hz, 20 V (rms) source is connected to a resistance of 100 Ω, an inductance of 0.1803 H and a capacitance of 10 μF all in series. The time in which the resistance (heat capacity 2 J/^{o}C) will get heated by 10°C. (assume no loss of heat to the surroudnings) is close to :

Open solution

(A) 348 s (B) 418 s (C) 245 s (D) 365 s

Q88. JEE Main 2020 (Online) 2nd September Evening Slot

An inductance coil has an impedance of 100 Ω. When an AC signal of frequency 1000 Hz is applied to the coil, the applied voltage leads the current by 45°. The self-inductance of the coil is

Open solution

(A) 6.7×10^{-7} H (B) 1.1×10^{-1} H (C) 5.5×10^{-5} H (D) 1.1×10^{-2} H

Q89. JEE Main 2020 (Online) 9th January Evening Slot

In LC circuit the inductance L = 40 mH and capacitance C = 100 μF. If a voltage V(t) = 10sin(314t) is applied to the circuit, the current in the circuit is given as :

Open solution

(A) 0.52 cos 314 t (B) 5.2 cos 314 t (C) 0.52 sin 314 t (D) 10 cos 314 t

Q90. JEE Main 2020 (Online) 7th January Evening Slot

An emf of 20 V is applied at time t = 0 to a circuit containing in series 10 mH inductor and 5 Ω resistor. The ratio of the currents at time t = ∞ and at t = 40 s is close to : (Take e^2 = 7.389)

Open solution

(A) 1.06 (B) 0.84 (C) 1.15 (D) 1.46

Q91. JEE Main 2020 (Online) 7th January Morning Slot

A LCR circuit behaves like a damped harmonic oscillator. Comparing it with a physical spring-mass damped oscillator having damping constant 'b', the correct equivalence would be:

Open solution

(A) L $\leftrightarrow$ k, C $\leftrightarrow$ b, R $\leftrightarrow$ m

(B) L $\leftrightarrow$ m, C $\leftrightarrow$ k, R $\leftrightarrow$ b

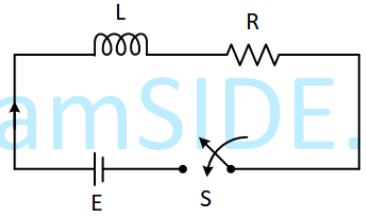
(C) L $\leftrightarrow$ m, C $\leftrightarrow \frac{1}{k}$, R $\leftrightarrow$ b

(D) L $\leftrightarrow \frac{1}{b}$, C $\leftrightarrow \frac{1}{m}$, R $\leftrightarrow \frac{1}{k}$

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Q92. JEE Main 2019 (Online) 12th April Evening Slot

Consider the LR circuit shown in the figure. If the switch S is closed at t = 0 then the amount of charge that passes through the battery between t = 0 and t = $\frac{L}{R}$ is :



ExamSIDE.com

Open solution

(A) $\frac{2.7EL}{R^2}$ (B) $\frac{EL}{2.7R^2}$ (C) $\frac{7.3EL}{R^2}$ (D) $\frac{EL}{7.3R^2}$

Q93. JEE Main 2019 (Online) 10th April Evening Slot

A coil of self inductance 10 mH and resistance 0.1 Ω is connected through a switch to a battery of internal resistance 0.9 Ω. After the switch is closed, the time taken for the current to attain 80% of the saturation value is: [take ln 5 = 1.6]

Open solution

(A) 0.324 s (B) 0.002 s (C) 0.103 s (D) 0.016 s

Q94. JEE Main 2019 (Online) 10th April Morning Slot

A transformer consisting of 300 turns in the primary and 150 turns in the secondary gives output power of 2.2 kW. If the current in the secondary coil is 10A, then the input voltage and current in the primary coil are :

Open solution

(A) 220 V and 20 A (B) 220 V and 10A (C) 440 V and 5A (D) 440 V and 20 A

Q95. JEE Main 2019 (Online) 8th April Evening Slot

A circuit connected to an ac source of emf e = $e_0$sin(100t) with t in seconds, gives a phase difference of $\pi/4$ between the emf e and current i. Which of the following circuits will exhibit this ?

Open solution

(A) RC circuit with R = 1 kΩ and C = 1μF

(B) RL circuit with R = 1kΩ and L = 1mH

(C) RC circuit with R = 1kΩ and C = 10 μF

(D) RL circuit with R = 1 kΩ and L = 10 mH

Q96. JEE Main 2019 (Online) 8th April Morning Slot

An alternating voltage v(t) = 220 sin 100 πt volt is applied to a purely resistance load of 50Ω . The time taken for the current to rise from half of the peak value to the peak value is :

Open solution

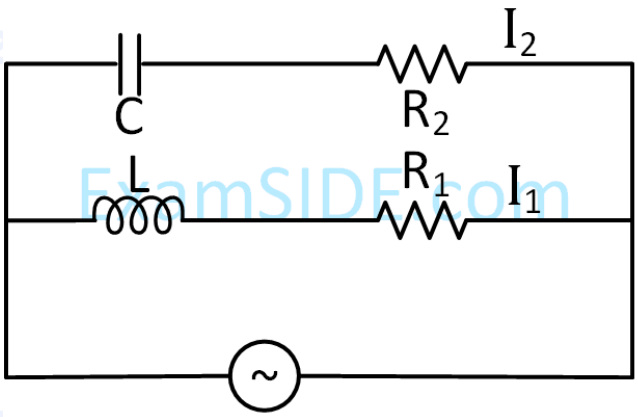
(A) 5 ms (B) 2.2 ms (C) 3.3 ms (D) 7.2 ms

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Q97. JEE Main 2019 (Online) 12th January Evening Slot

Open solution

In the above circuit, C = $\frac{\sqrt{3}}{2}\mu$F, R_2 = 20 Ω, L = $\frac{\sqrt{3}}{10}$ H and R_1 = 10 Ω. Current in L-R_1 path is I_1 and in C-R_2 path it is I_2 . The voltage of A.C. source is given by, V = 200 $\sqrt{2}$ sin (100 t) volts . The phase difference between I_1 and I_2 is :

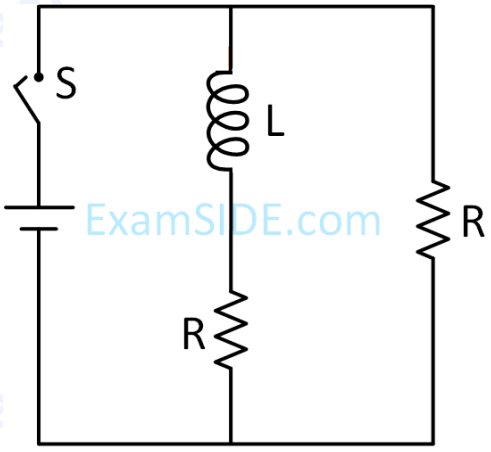


(A) 150° (B) 90° (C) 30° (D) 0°

Q98. JEE Main 2019 (Online) 12th January Morning Slot

Open solution

In the figure shown, a circuit contains two identical resistors with resistance R = 5Ω and an inductance with L = 2mH. An ideal battery of 15 V is connected in the circuit. What will be the current through the battery long after the switch is closed ?



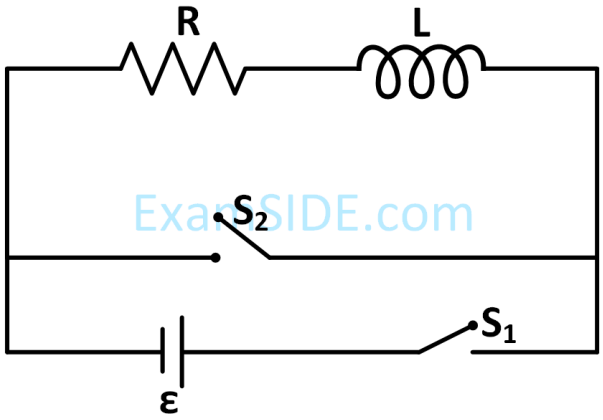
(A) 6 A (B) 7.5 A (C) 3 A (D) 5.5 A

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Q99. JEE Main 2019 (Online) 11th January Morning Slot

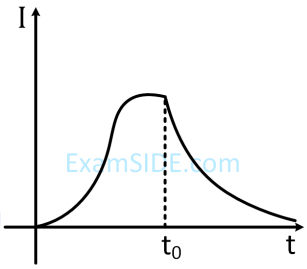
Open solution

In the circuit shown,

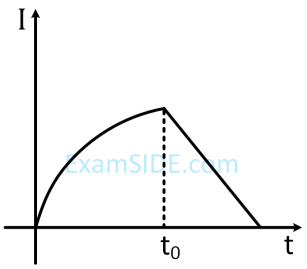


the switch S_1 is closed at time t = 0 and the switch S_2 is kept open. At some later time(t_0), the switch S_1 is opened and S_2 is closed. The behavior of the current I as a function of time 't' is given by :

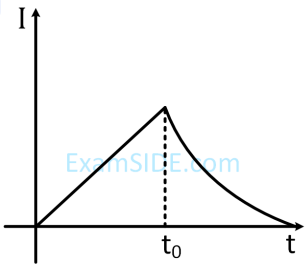
(A)



(B)

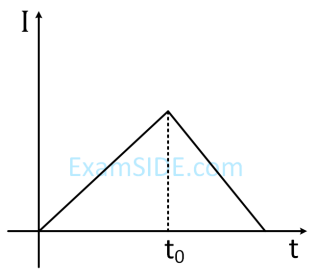


(C)



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(D)



Q100. JEE Main 2019 (Online) 9th January Evening Slot

A power transmission line feeds input power at 2300 V to a step down transformer with its primary windings having 4000 turns. The output power is delivered at 230 V by the transformer. If the current in the primary of the transformer is 5A and its efficiency is 90%, the output current would be :

Open solution

(A) 50 A (B) 45 A (C) 35 A (D) 25 A

Q101. JEE Main 2019 (Online) 9th January Evening Slot

A series AC circuit containing an inductor (20 mH), a capacitor (120 μF) and a resistor (60 Ω) is driven by an AC source of 24V/50 Hz. The energy dissipated in the circuit in 60 s is :

Open solution

(A) 5.65×10^2J (B) 2.26×10^3J (C) 5.17×10^2 J (D) 3.39×10^3 J

Q102. JEE Main 2018 (Online) 16th April Morning Slot

A power transmission line feeds input power at 2300 V to a step down transformer with its primary windings having 4000 turns, giving the output power at 230 V. If the current in the primary of the transformer is 5 A, and its efficiency is 90%, the output current would be ;

Open solution

(A) 50 A (B) 45 A (C) 25 A (D) 20 A

Q103. JEE Main 2018 (Offline)

In an a.c. circuit, the instantaneous e.m.f. and current are given by
e = 100 sin 30 t
i = 20 sin $\left(30t - \frac{\pi}{4}\right)$
In one cycle of a.c., the average power consumed by the circuit and the wattless current are, respectively

Open solution

(A) 50, 0 (B) 50, 10 (C) $\frac{1000}{\sqrt{2}}, 10$ (D) $\frac{50}{\sqrt{2}}$

Q104. JEE Main 2018 (Online) 15th April Morning Slot

An ideal capacitor of capacitance $0.2\,\mu F$ is charged to a potential difference of $10\,V$. The charging battery is then disconnected. The capacitor is then connected to an ideal inductor of self inductance $0.5\,mH$. The current at a time when the potential difference across the capacitor is $5\,V$, is :

Open solution

(A) $0.34\,A$ (B) $0.25\,A$ (C) $0.17\,A$ (D) $0.15\,A$

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Q105. JEE Main 2017 (Online) 9th April Morning Slot

Open solution

A sinusoidal voltage of peak value 283 V and angular frequency 320/s is applied to a series LCR circuit. Given that R=5 Ω, L=25 mH and C=1000 μF. The total impedance, and phase difference between the voltage across the source and the current will respectively be :

(A) 10 Ω and $\tan^{-1}\left(\frac{5}{3}\right)$

(B) 7 Ω and 45°

(C) 10 Ω and $\tan^{-1}\left(\frac{8}{3}\right)$

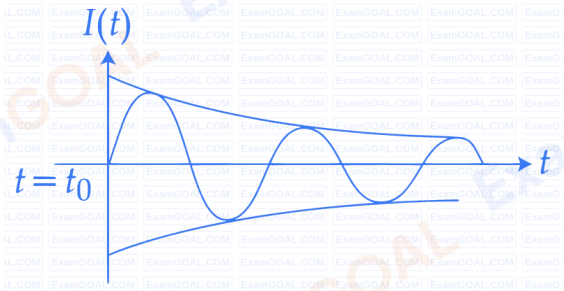
(D) 7 Ω and $\tan^{-1}\left(\frac{5}{3}\right)$

Q106. JEE Main 2016 (Online) 9th April Morning Slot

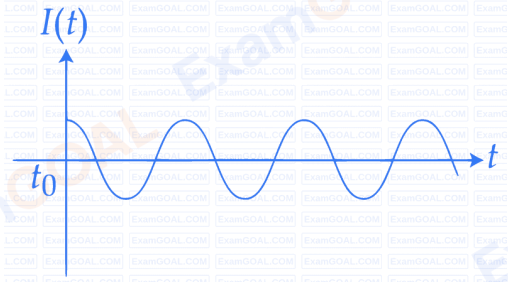
Open solution

A series *LR* circuit is connected to a voltage source with $V(t) = V_0 \sin\Omega t$. After very large time, current *I(t)* behaves as ($t_0 >> \frac{L}{R}$) :

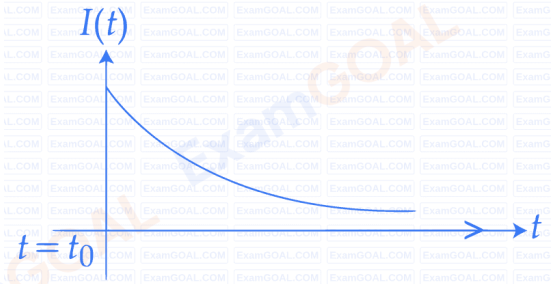
(A)



(B)



(C)



(D)



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Q107. JEE Main 2016 (Offline)

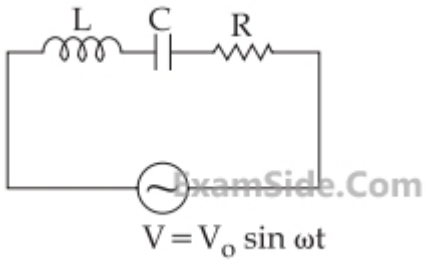
An arc lamp requires a direct current of 10 A at 80 V to function. If it is connected to a 220 V (rms), 50 Hz AC supply, the series inductor needed for it to work is close to :

(A) 0.044 H (B) 0.065 H (C) 80 H (D) 0.08 H

Open solution

Q108. JEE Main 2015 (Online) 11th April Morning Slot

For the LCR circuit, shown here, the current is observed to lead the applied voltage. An additional capacitor C' , when joined with the capacitor C present in the circuit , makes the power factor of the circuit unity. The capacitor C', must have been connected in :



(A) series with C and has a magnitude

$\frac{1-\omega^2 LC}{\omega^2 L}$

(B) series with C and has a magnitude

$\frac{C}{(\omega^2 LC-1)}$.

(C) parallel with C and has a magnitude

$\frac{C}{(\omega^2 LC-1)}$

(D) parallel with C and has a magnitude

$\frac{1-\omega^2 LC}{\omega^2 L}$

Open solution

Q109. JEE Main 2015 (Online) 11th April Morning Slot

The AC voltage across a resistance can be measured using a :

(A) potentiometer (B) moving coil galvanometer

(C) moving magnet galvanometer (D) hot wire voltmeter

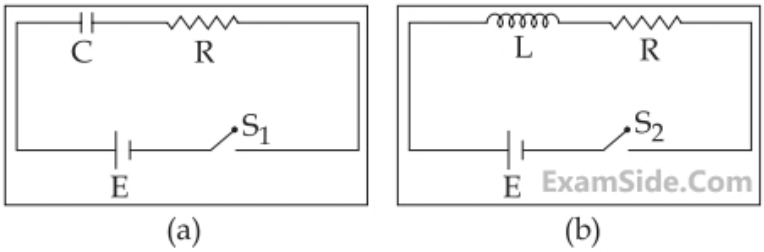
Open solution

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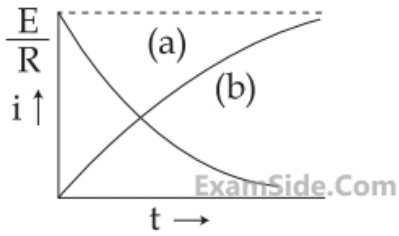
Q110. JEE Main 2015 (Online) 10th April Morning Slot

Open solution

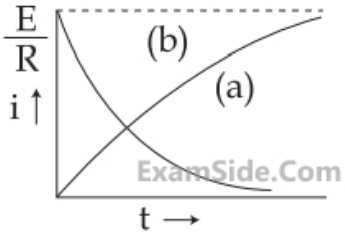
In the circuits (a) and (b) switches S_1 and S_2 are closed at t = 0 and are kept closed for a long time. The variation of currents in the two circuits for $t \geq 0$ are roughly shown by (figures are schematic and not drawn to scale) :



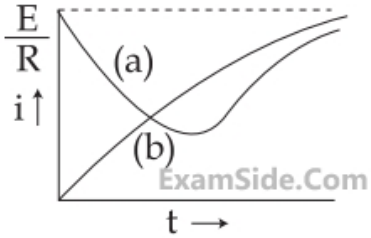
(A)



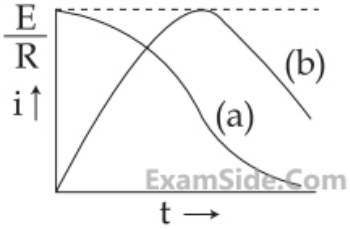
(B)



(C)



(D)

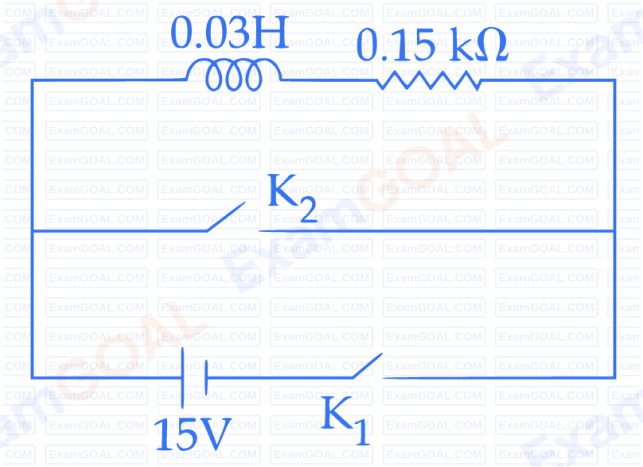


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Q111. JEE Main 2015 (Offline)

Open solution

An inductor $(L = 0.03\ H)$ and a resistor $(R = 0.15\ k\Omega)$ are connected in series to a battery of $15V$ EMF in a circuit shown below. The key K_1 has been kept closed for a long time. Then at $t = 0$, K_1 is opened and key K_2 is closed simultaneously. At $t = 1\ ms$, the current in the circuit will be : $(e^5 \cong 150)$



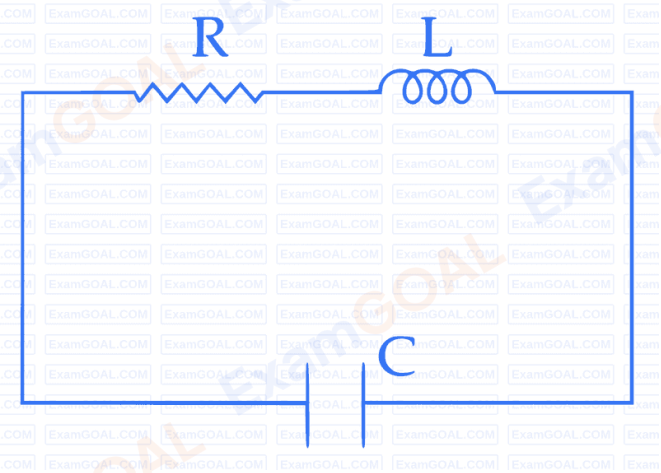
(A) $6.7\ mA$ (B) $0.67\ mA$ (C) $100\ mA$ (D) $67\ mA$

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Q112. JEE Main 2015 (Offline)

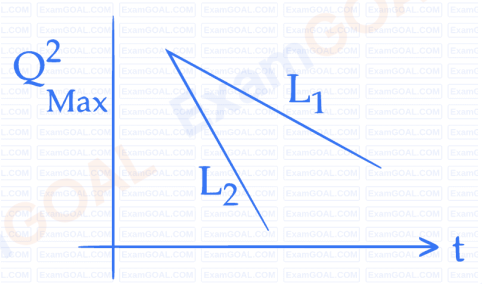
Open solution

An LCR circuit is equivalent to a damped pendulum. In an LCR circuit the capacitor is charged to Q_0 and then connected to the L and R as shown below :

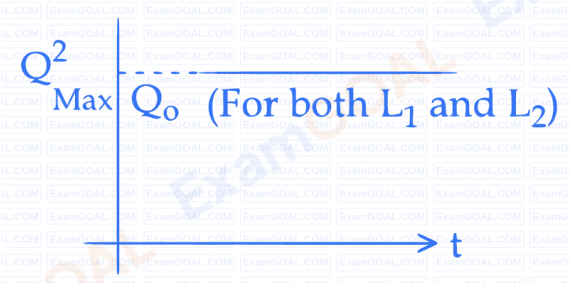


If a student plots graphs of the square of maximum charge (Q^2_{Max}) on the capacitor with time(t) for two different values L_1 and L_2 $(L_1 > L_2)$ of L then which of the following represents this graph correctly ?
(plots are schematic and not drawn to scale)

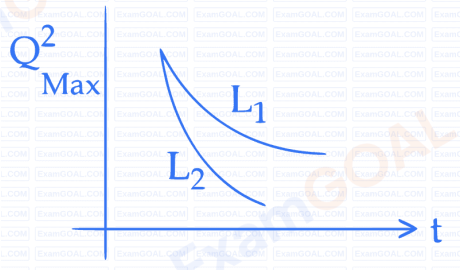
(A)



(B)

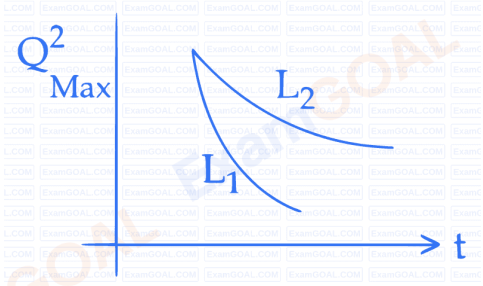


(C)



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(D)



Q113. JEE Main 2014 (Online) 12th April Morning Slot

A sinusoidal voltage V(t) = 100 sin (500t) is applied across a pure inductance of L = 0.02 H. The current through the coil is :

(A) 10 cos (500t) (B) − 10 cos (500t) (C) 10 sin (500t) (D) − 10 sin (500t)

Open solution

Q114. JEE Main 2014 (Online) 9th April Morning Slot

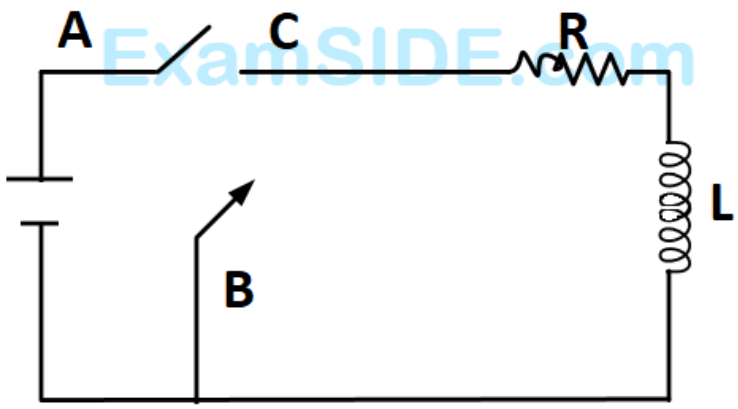
When the rms voltages V_L, V_C and V_R are measured respectively across the inductor L, the capacitor C and the resistor R in a series LCR circuit connected to an AC source, it is found that the ratio $V_L : V_C : V_R = 1 : 2 : 3$. If the rms voltage of the AC source is 100 V, then V_R is close to :

(A) 50 V (B) 70 V (C) 90 V (D) 100 V

Open solution

Q115. JEE Main 2014 (Offline)

In the circuit shown here, the point $'C'$ is kept connected to point $'A'$ till the current flowing through the circuit becomes constant. Afterward, suddenly, point $'C'$ is disconnected from point $'A'$ and connected to point $'B'$ at time $t = 0$. Ratio of the voltage across resistance and the inductor at $t = L/R$ will be equal to :



(A) $\frac{e}{1-e}$ (B) 1 (C) -1 (D) $\frac{1-e}{e}$

Open solution

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Q116. JEE Main 2013 (Online) 25th April Morning Slot

When resonance is produced in a series LCR circuit, then which of the following is not correct?

(A) Current in the circuit is in phase with the applied voltage.

(B) Inductive and capacitive reactances are equal.

(C) If R is reduced, the voltage across capacitor will increase.

(D) Impedance of the circuit is maximum.

Open solution

Q117. JEE Main 2013 (Online) 25th April Morning Slot

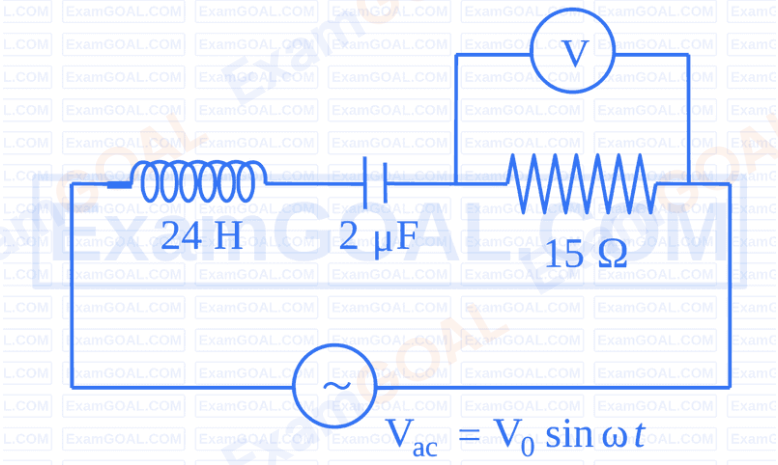
A series LR circuit is connected to an ac source of frequency ω and the inductive reactance is equal to 2R. A capacitance of capacitive reactance equal to R is added in series with L and R. The ratio of the new power factor to the old one is :

(A) $\sqrt{\frac{2}{3}}$ (B) $\sqrt{\frac{2}{5}}$ (C) $\sqrt{\frac{3}{2}}$ (D) $\sqrt{\frac{5}{2}}$

Open solution

Q118. JEE Main 2013 (Online) 22th April Morning Slot

An LCR circuit as shown in the figure is connected to a voltage source V_{ac} whose frequency can be varied. The frequency, at which the voltage across the resistor is maximum, is :



(A) 902 Hz (B) 143 Hz (C) 23 Hz (D) 345 Hz

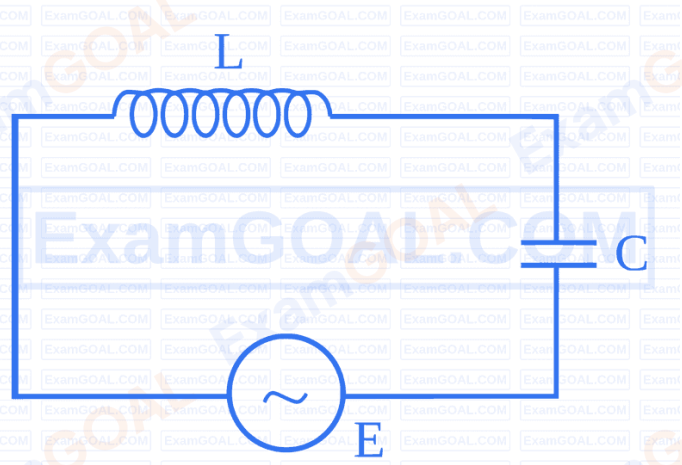
Open solution

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Q119. JEE Main 2013 (Online) 9th April Morning Slot

In the circuit shown here, the voltage across L and C are respectively 300 V and 400 V. The voltage E of the ac source is :

Open solution



(A) 400 Volt (B) 500 Volt (C) 100 Volt (D) 700 Volt

Q120. AIEEE 2012 (Online) 19th May

A generator has armature resistance of $0.1\ \Omega$ and develops an induced emf of $120\ \text{V}$ when driven at its rated speed. Its terminal voltage when a current of $50\ \text{A}$ is being drawn is

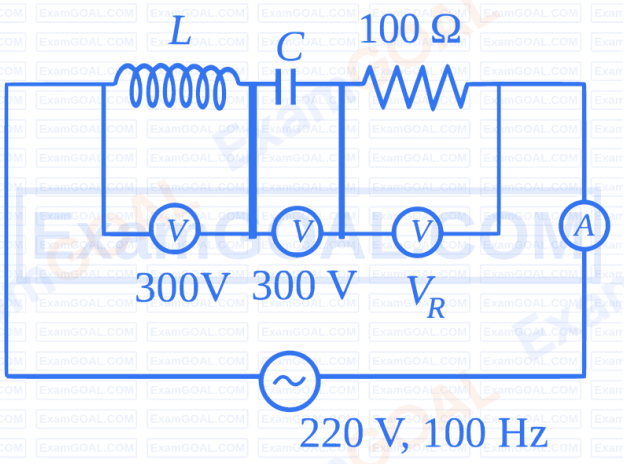
(A) $120\ \text{V}$ (B) $5\ \text{V}$ (C) $115\ \text{V}$ (D) $70\ \text{V}$

Open solution

Q121. AIEEE 2012 (Online) 7th May

In an LCR circuit shown in the following figure, what will be the readings of the voltmeter across the resistor and ammeter if an a.c. source of $220\ \text{V}$ and $100\ \text{Hz}$ is connected to it as shown?

Open solution



(A) $800\ \text{V},\ 8\ \text{A}$ (B) $110\ \text{V},\ 1.1\ \text{A}$

(C) $300\ \text{V},\ 3\ \text{A}$ (D) $220\ \text{V},\ 2.2\ \text{A}$

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Q122. AIEEE 2011

A fully charged capacitor C with initial charge q_0 is connected to a coil of self inductance L at $t = 0$. The time at which the energy is stored equally between the electric and the magnetic fields is :

(A) $\frac{\pi}{4}\sqrt{LC}$ (B) $2\pi\sqrt{LC}$ (C) $\sqrt{LC}$ (D) $\pi\sqrt{LC}$

Open solution

Q123. AIEEE 2010

In a series LCR circuit $R = 200\Omega$ and the voltage and the frequency of the main supply are $220V$ and $50\ Hz$ respectively. On taking out the capacitance from the circuit the current lags behind the voltage by 30°. On taking out the inductor from the circuit the current leads the voltage by 30°. The power dissipated in the LCR circuit is

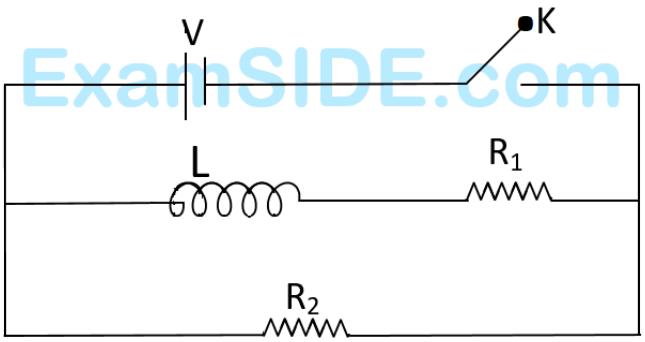
(A) $305\ W$ (B) $210\ W$ (C) zero (D) $242\ W$

Open solution

Q124. AIEEE 2010

In the circuit shown below, the key K is closed at $t = 0$. The current through the battery is

Open solution



(A) $\frac{VR_1R_2}{\sqrt{R_1^2+R_2^2}}$ at $t = 0$ and $\frac{V}{R_2}$ at $t = \infty$

(B) $\frac{V}{R_2}$ at $t = 0$ and $\frac{V(R_1+R_2)}{R_1R_2}$ at $t = \infty$

(C) $\frac{V}{R_2}$ at $t = 0$ and $\frac{VR_1R_2}{\sqrt{R_1^2+R_2^2}}$ at $t = \infty$

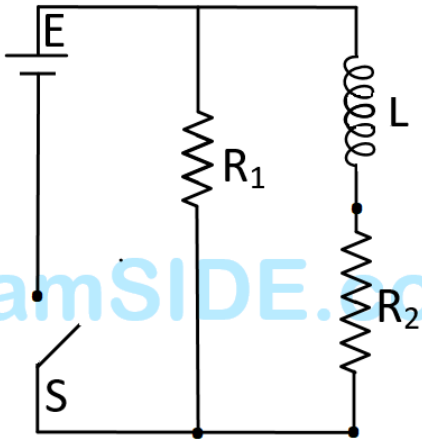
(D) $\frac{V(R_1+R_2)}{R_1R_2}$ at $t = 0$ and $\frac{V}{R_2}$ at $t = \infty$

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Q125. AIEEE 2009

An inductor of inductance $L = 400\,mH$ and resistors of resistance $R_1 = 2\Omega$ and $R_2 = 2\Omega$ are connected to a battery of $emf\ 12\ V$ as shown in the figure. The internal resistance of the battery is negligible. The switch S is closed at $t = 0$. The potential drop across L as a function of time is :

Open solution



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(A) $\frac{12}{t}e^{-3t}V$ (B) $6\left(1-e^{-t/0.2}\right)V$

(C) $12e^{-5t}V$ (D) $6e^{-5t}V$

Q126. AIEEE 2007

In an $a.c.$ circuit the voltage applied is $E = E_0\ \sin\ \omega t$. The resulting current in the circuit is $I = I_0 \sin\left(\omega t - \frac{\pi}{2}\right)$. The power consumption in the circuit is given by

Open solution

(A) $P = \sqrt{2}E_0I_0$ (B) $P = \frac{E_0I_0}{\sqrt{2}}$

(C) $P = zero$ (D) $P = \frac{E_0I_0}{2}$

Q127. AIEEE 2006

In a series resonant LCR circuit, the voltage across R is 100 volts and $R = 1\,k\Omega$ with $C = 2\mu F$. The resonant frequency ω is $200\ rad/s$. At resonance the voltage across L is

Open solution

(A) $2.5 \times 10^{-2}V$ (B) $40\ V$ (C) $250\ V$ (D) $4 \times 10^{-3}V$

Q128. AIEEE 2006

In an AC generator, a coil with N turns, all of the same area A and total resistance R, rotates with frequency ω in a magnetic field B. The maximum value of emf generated in the coil is

Open solution

(A) $N.A.B.R.\omega$ (B) $N.A.B$ (C) $N.A.B.R.$ (D) $N.A.B.\omega$

Q129. AIEEE 2005

The self inductance of the motor of an electric fan is $10\ H$. In order to impart maximum power at $50\ Hz$, it should be connected to a capacitance of

Open solution

(A) $8\mu F$ (B) $4\mu F$ (C) $2\mu F$ (D) $1\mu F$

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Q130. AIEEE 2005

The phase difference between the alternating current and emf is $\frac{\pi}{2}$. Which of the following cannot be the constituent of the circuit?

(A) R, L (B) C alone (C) L alone (D) L, C

Open solution

Q131. AIEEE 2005

A circuit has a resistance of $12\ ohm$ and an impedance of $15\ ohm$. The power factor of the circuit will be

(A) 0.4 (B) 0.8 (C) 0.125 (D) 1.25

Open solution

Q132. AIEEE 2004

Alternating current can not be measured by $D.C.$ ammeter because

(A) Average value of current for complete cycle is zero

(B) $A.C.$ Changes direction

(C) $A.C.$ can not pass through $D.C.$ Ammeter

(D) $D.C.$ Ammeter will get damaged.

Open solution

Q133. AIEEE 2004

In an LCR series $a.c.$ circuit, the voltage across each of the components, L, C and R is $50V$. The voltage across the $L.C$ combination will be :

(A) $100V$ (B) $50\sqrt{2}$ (C) $50\ V$ (D) $0\ V$ (zero)

Open solution

Q134. AIEEE 2004

In a LCR circuit capacitance is changed from C to $2\,C$. For the resonant frequency to remain unchaged, the inductance should be changed from L to

(A) $L/2$ (B) $2L$ (C) $4L$ (D) $L/4$

Open solution

Q135. AIEEE 2003

In an oscillating LC circuit the maximum charge on the capacitor is Q. The charge on the capacitor when the energy is stored equally between the electric and magnetic field is

(A) $\frac{Q}{2}$ (B) $\frac{Q}{\sqrt{3}}$ (C) $\frac{Q}{\sqrt{2}}$ (D) Q

Open solution

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Q136. AIEEE 2003

The core of any transformer is laminated so as to

(A) reduce the energy loss due to eddy currents

(B) make it light weight

(C) make it robust and strong

(D) increase the secondary voltage

Open solution

Q137. AIEEE 2002

The power factor of AC circuit having resistance (R) and inductance (L) connected in series and an angular velocity ω is

(A) $R/\omega L$

(B) $R/\left(R^2+\omega^2L^2\right)^{1/2}$

(C) $\omega L/R$

(D) $R/\left(R^2-\omega^2L^2\right)^{1/2}$

Open solution

Q138. AIEEE 2002

In a transformer, number of turns in the primary coil are 140 and that in the secondary coil are 280. If current in primary coil is $4A$, then that in the secondary coil is

(A) $4A$

(B) $2A$

(C) $6A$

(D) $10A$

Open solution

Numerical 57 questions

Q139. JEE Main 2026 (Online) 5th April Evening Shift

A series LCR circuit with $R=20\Omega, L=1.6\text{H}$ and $C=40\mu\text{ F}$ is connected to a variable frequency a.c. source. The inductive reactance at resonant frequency is ____ Ω.

Open solution

Q140. JEE Main 2026 (Online) 4th April Morning Shift

An inductor of 10 mH , capacitor of $0.1\mu\text{ F}$ and a resistor of 100Ω are connected in series across an $a.c$ power supply $220\text{ V}, 70\text{ Hz}$. The power factor of the given circuit is 0.5 . The difference in the inductive reactance and capacitance reactance is $\sqrt{3}\alpha\Omega$. The value of α is ____ .

Open solution

Q141. JEE Main 2026 (Online) 28th January Evening Shift

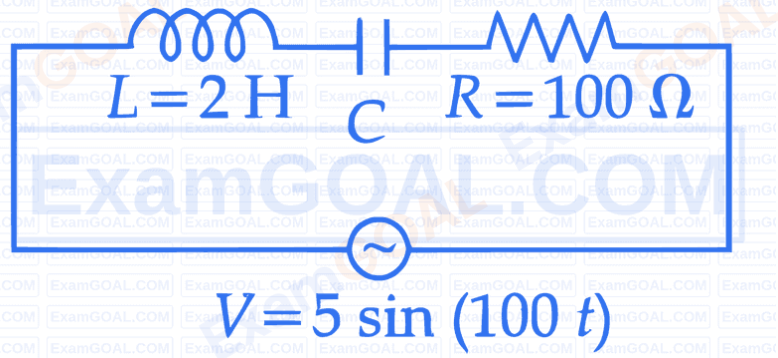
An inductor stores 16 J of magnetic field energy and dissipates 32 W of thermal energy due to its resistance when an a.c. current of 2 A (rms) and frequency 50 Hz flows through it. The ratio of inductive reactance to its resistance is _____. $(\pi=3.14)$

Open solution

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Q142. JEE Main 2026 (Online) 23rd January Morning Shift

Using a variable frequency a.c. voltage source the maximum current measured in the given LCR circuit is 50 mA for $V = 5\sin(100t)$ The values of L and R are shown in the figure. The capacitance of the capacitor (C) used is ____ μF.



Open solution

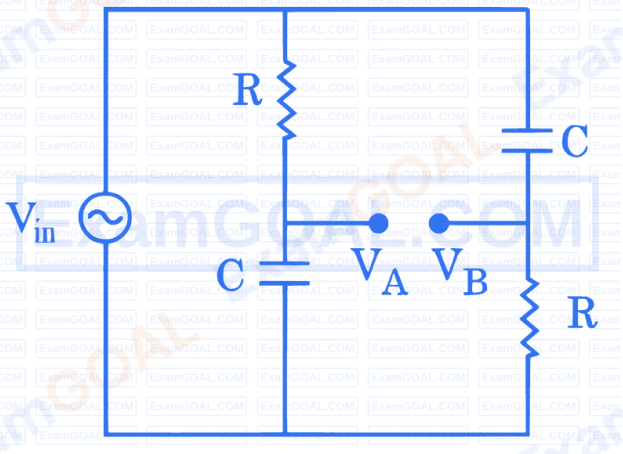
Q143. JEE Main 2025 (Online) 7th April Evening Shift

An inductor of reactance 100Ω, a capacitor of reactance 50Ω, and a resistor of resistance 50Ω are connected in series with an AC source of $10\ \text{V}, 50\ \text{Hz}$. Average power dissipated by the circuit is _____ W.

Open solution

Q144. JEE Main 2025 (Online) 7th April Morning Shift

For ac circuit shown in figure, $\text{R} = 100\text{k}\Omega$ and $\text{C} = 100\text{pF}$ and the phase difference between V_{in} and $(\text{V}_\text{B} - \text{V}_\text{A})$ is 90°. The input signal frequency is 10^xrad/sec, where ' x ' is _____ .



Open solution

Q145. JEE Main 2025 (Online) 4th April Evening Shift

An inductor of self inductance 1 H is connected in series with a resistor of 100π ohm and an ac supply of 100π volt, 50 Hz . Maximum current flowing in the circuit is _____ A.

Open solution

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Q146. JEE Main 2025 (Online) 23rd January Evening Shift

In a series LCR circuit, a resistor of 300Ω, a capacitor of 25 nF and an inductor of 100 mH are used. For maximum current in the circuit, the angular frequency of the ac source is _____ $\times 10^4$ radians s^{-1}

Open solution

Q147. JEE Main 2024 (Online) 9th April Evening Shift

A capacitor of reactance $4\sqrt{3}\Omega$ and a resistor of resistance 4Ω are connected in series with an ac source of peak value $8\sqrt{2}$ V. The power dissipation in the circuit is _____ W.

Open solution

Q148. JEE Main 2024 (Online) 9th April Morning Shift

When a coil is connected across a 20 V dc supply, it draws a current of 5 A. When it is connected across 20 V, 50 Hz ac supply, it draws a current of 4 A. The self inductance of the coil is _____ mH. (Take $\pi = 3$)

Open solution

Q149. JEE Main 2024 (Online) 8th April Evening Shift

An alternating emf $E = 110\sqrt{2}\sin 100t$ volt is applied to a capacitor of $2\mu F$, the rms value of current in the circuit is _____ mA.

Open solution

Q150. JEE Main 2024 (Online) 6th April Evening Shift

For a given series LCR circuit it is found that maximum current is drawn when value of variable capacitance is 2.5 nF. If resistance of 200Ω and 100 mH inductor is being used in the given circuit. The frequency of ac source is _____ $\times 10^3$ Hz (given $\pi^2 = 10$)

Open solution

Q151. JEE Main 2024 (Online) 6th April Morning Shift

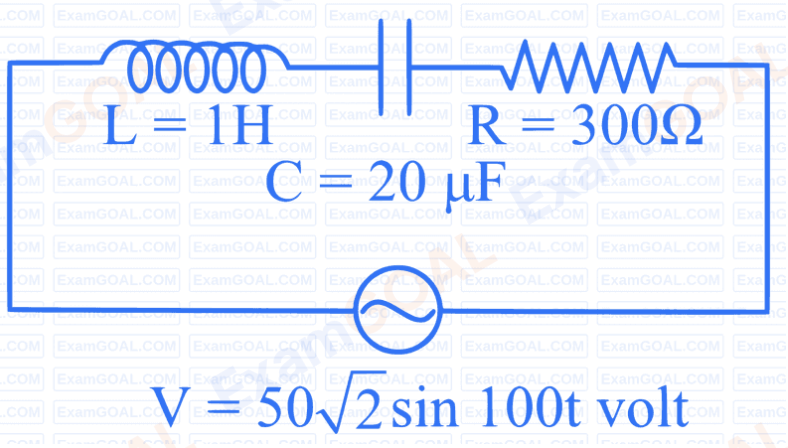
When a dc voltage of 100 V is applied to an inductor, a dc current of 5 A flows through it. When an ac voltage of 200 V peak value is connected to inductor, its inductive reactance is found to be $20\sqrt{3}\Omega$. The power dissipated in the circuit is _____ W.

Open solution

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Q152. JEE Main 2024 (Online) 5th April Morning Shift

An ac source is connected in given series LCR circuit. The rms potential difference across the capacitor of $20\mu F$ is _____ V.



Open solution

Q153. JEE Main 2024 (Online) 4th April Morning Shift

An alternating current at any instant is given by $i = [6 + \sqrt{56}\sin(100\pi t + \pi/3)]$ A. The rms value of the current is _____ A.

Open solution

Q154. JEE Main 2024 (Online) 30th January Evening Shift

A power transmission line feeds input power at $2.3\ \mathrm{kV}$ to a step down transformer with its primary winding having 3000 turns. The output power is delivered at $230\ \mathrm{V}$ by the transformer. The current in the primary of the transformer is $5\ \mathrm{A}$ and its efficiency is 90%. The winding of transformer is made of copper. The output current of transformer is _____ A.

Open solution

Q155. JEE Main 2024 (Online) 27th January Evening Shift

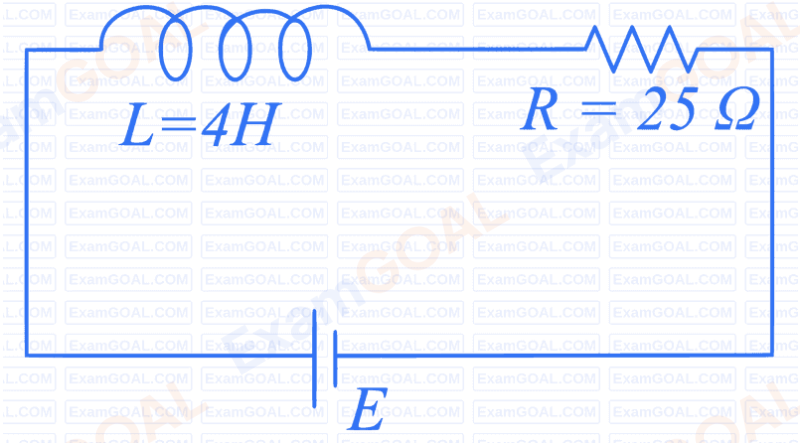
A series LCR circuit with $\mathrm{L} = \frac{100}{\pi}\mathrm{mH}, \mathrm{C} = \frac{10^{-3}}{\pi}\mathrm{F}$ and $\mathrm{R} = 10\Omega$, is connected across an ac source of $220\ \mathrm{V}, 50\ \mathrm{Hz}$ supply. The power factor of the circuit would be _____.

Open solution

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Q156. JEE Main 2023 (Online) 13th April Morning Shift

In the given figure, an inductor and a resistor are connected in series with a battery of emf E volt. $\frac{E^a}{2b}$ J/s represents the maximum rate at which the energy is stored in the magnetic field (inductor). The numerical value of $\frac{b}{a}$ will be _____.



Open solution

Q157. JEE Main 2023 (Online) 11th April Evening Shift

A coil has an inductance of $2\mathrm{H}$ and resistance of $4\,\Omega$. A $10\ \mathrm{V}$ is applied across the coil. The energy stored in the magnetic field after the current has built up to its equilibrium value will be _____ $\times 10^{-2}\ \mathrm{J}$.

Open solution

Q158. JEE Main 2023 (Online) 8th April Morning Shift

An oscillating LC circuit consists of a $75\ \mathrm{mH}$ inductor and a $1.2\ \mu\mathrm{F}$ capacitor. If the maximum charge to the capacitor is $2.7\ \mu\mathrm{C}$. The maximum current in the circuit will be _____ mA

Open solution

Q159. JEE Main 2023 (Online) 6th April Morning Shift

An ideal transformer with purely resistive load operates at $12\ \mathrm{kV}$ on the primary side. It supplies electrical energy to a number of nearby houses at $120\ \mathrm{V}$. The average rate of energy consumption in the houses served by the transformer is 60 kW. The value of resistive load (Rs) required in the secondary circuit will be _____ $\mathrm{m}\Omega$.

Open solution

Q160. JEE Main 2023 (Online) 1st February Evening Shift

A square shaped coil of area $70\ \mathrm{cm}^2$ having 600 turns rotates in a magnetic field of $0.4\ \mathrm{wbm}^{-2}$, about an axis which is parallel to one of the sides of the coil and perpendicular to the direction of field. If the coil completes 500 revolutions in a minute, the instantaneous emf when the plane of the coil is inclined at 60° with the field, will be _____ V. (Take $\pi = \frac{22}{7}$)

Open solution

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Q161. JEE Main 2023 (Online) 1st February Morning Shift

A series LCR circuit is connected to an ac source of $220\ \mathrm{V}, 50\ \mathrm{Hz}$. The circuit contain a resistance $\mathrm{R} = 100\ \Omega$ and an inductor of inductive reactance $\mathrm{X_L} = 79.6\ \Omega$. The capacitance of the capacitor needed to maximize the average rate at which energy is supplied will be _____ $\mu\mathrm{F}$.

Open solution

Q162. JEE Main 2023 (Online) 31st January Evening Shift

A series LCR circuit consists of $\mathrm{R} = 80\Omega, \mathrm{X_L} = 100\Omega$, and $\mathrm{X_C} = 40\Omega$. The input

voltage is 2500 $\cos(100\pi\mathrm{t})\mathrm{V}$. The amplitude of current, in the circuit, is _____ A.

Open solution

Q163. JEE Main 2023 (Online) 31st January Morning Shift

An inductor of $0.5\ \mathrm{mH}$, a capacitor of $20\ \mu\mathrm{F}$ and resistance of $20\ \Omega$ are connected in series with a $220\ \mathrm{V}$ ac source. If the current is in phase with the emf, the amplitude of current of the circuit is $\sqrt{x}$ A. The value of x is _____

Open solution

Q164. JEE Main 2023 (Online) 30th January Evening Shift

In an ac generator, a rectangular coil of 100 turns each having area $14 \times 10^{-2}\ \mathrm{m}^2$ is rotated at $360\ \mathrm{rev/min}$ about an axis perpendicular to a uniform magnetic field of magnitude $3.0\ \mathrm{T}$. The maximum value of the emf produced will be _____ V.

(Take $\pi = \frac{22}{7}$)

Open solution

Q165. JEE Main 2023 (Online) 29th January Evening Shift

An inductor of inductance 2 $\mu\mathrm{H}$ is connected in series with a resistance, a variable capacitor and an AC source of frequency 7 kHz. The value of capacitance for which maximum current is drawn into the circuit $\frac{1}{x}\mathrm{F}$, where the value of x is _____.

(Take $\pi = \frac{22}{7}$)

Open solution

Q166. JEE Main 2023 (Online) 25th January Evening Shift

A series LCR circuit is connected to an AC source of 220 V, 50 Hz. The circuit contains a resistance R = 80Ω, an inductor of inductive reactance $\mathrm{X_L} = 70\Omega$, and a capacitor of capacitive reactance $\mathrm{X_C} = 130\Omega$. The power factor of circuit is $\frac{x}{10}$. The value of x is :

Open solution

Q167. JEE Main 2023 (Online) 25th January Morning Shift

An LCR series circuit of capacitance 62.5 nF and resistance of 50 Ω, is connected to an A.C. source of frequency 2.0 kHz. For maximum value of amplitude of current in circuit, the value of inductance is _____ mH.

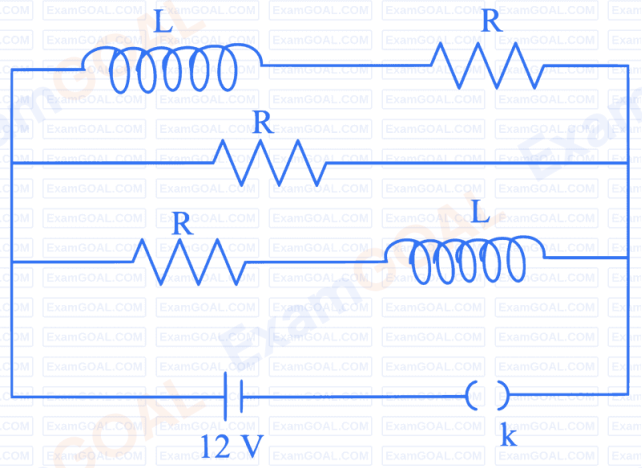
(Take $\pi^2 = 10$)

Open solution

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Q168. JEE Main 2023 (Online) 24th January Evening Shift

Three identical resistors with resistance R = 12Ω and two identical inductors with self inductance L = 5 mH are connected to an ideal battery with emf of 12 V as shown in figure. The current through the battery long after the switch has been closed will be _____ A.



Open solution

Q169. JEE Main 2022 (Online) 30th July Evening Shift

A sinusoidal voltage $V(t) = 200 \sin 2000t$ volt is applied to a series LCR circuit in which $L = 10\ \text{mH}$, $C = 25\ \mu\text{F}$ and $R = 100\ \Omega$. The impedance of the circuit is _____ Ω

Open solution

Q170. JEE Main 2022 (Online) 29th July Evening Shift

A capacitor of capacitance 500 μF is charged completely using a dc supply of 100 V. It is now connected to an inductor of inductance 50 mH to form an LC circuit. The maximum current in LC circuit will be _____ A.

Open solution

Q171. JEE Main 2022 (Online) 28th July Morning Shift

The frequencies at which the current amplitude in an LCR series circuit becomes $\frac{1}{\sqrt{2}}$ times its maximum value, are $212\,\text{rad}\,\text{s}^{-1}$ and $232\,\text{rad}\,\text{s}^{-1}$. The value of resistance in the circuit is $R = 5\,\Omega$. The self inductance in the circuit is _____ mH.

Open solution

Q172. JEE Main 2022 (Online) 27th July Morning Shift

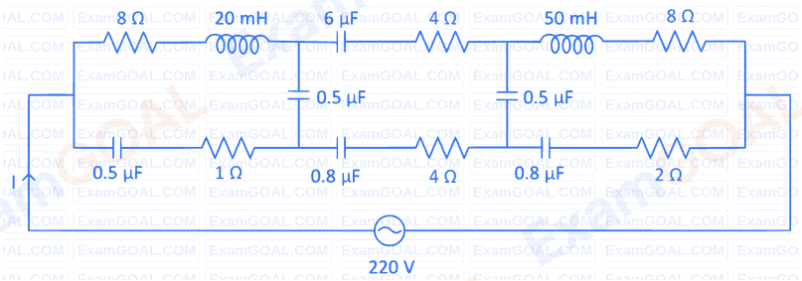
To light, a $50\ \text{W}, 100\ \text{V}$ lamp is connected, in series with a capacitor of capacitance $\frac{50}{\pi\sqrt{x}}\mu F$, with $200\ \text{V}, 50\ \text{Hz}$ AC source. The value of x will be _____.

Open solution

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Q173. JEE Main 2022 (Online) 26th July Morning Shift

The effective current I in the given circuit at very high frequencies will be _____ A.



Open solution

Q174. JEE Main 2022 (Online) 30th June Morning Shift

A series LCR circuit with $R = \frac{250}{11}\ \Omega$ and $X_L = \frac{70}{11}\ \Omega$ is connected across a 220 V, 50 Hz supply. The value of capacitance needed to maximize the average power of the circuit will be _____ μF. (Take : $\pi = \frac{22}{7}$)

Open solution

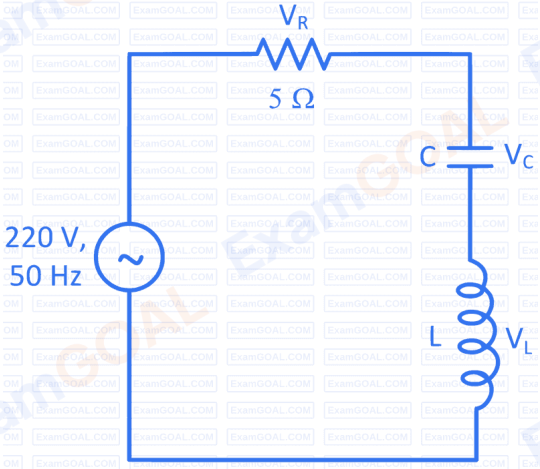
Q175. JEE Main 2022 (Online) 29th June Evening Shift

An inductor of 0.5 mH, a capacitor of 200 μF and a resistor of 2 Ω are connected in series with a 220 V ac source. If the current is in phase with the emf, the frequency of ac source will be _____ $\times 10^2$ Hz.

Open solution

Q176. JEE Main 2022 (Online) 28th June Evening Shift

In the given circuit, the magnitude of V_L and V_C are twice that of V_R. Given that f = 50 Hz, the inductance of the coil is $\frac{1}{K\pi}$ mH. The value of K is _____.



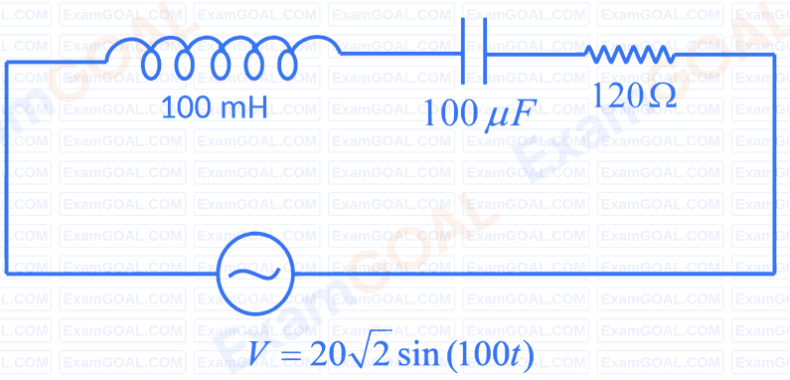
Open solution

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Q177. JEE Main 2022 (Online) 28th June Morning Shift

An AC source is connected to an inductance of 100 mH, a capacitance of 100 μF and a resistance of 120 Ω as shown in figure. The time in which the resistance having a thermal capacity 2 J/°C will get heated by 16°C is _____ s.

Open solution



Q178. JEE Main 2022 (Online) 28th June Morning Shift

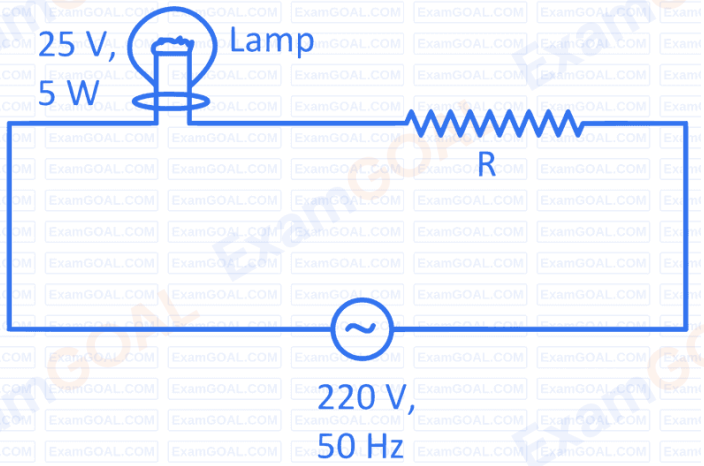
A telegraph line of length 100 km has a capacity of 0.01 μF/km and it carries an alternating current at 0.5 kilo cycle per second. If minimum impedance is required, then the value of the inductance that needs to be introduced in series is _____ mH. (if $\pi = \sqrt{10}$)

Open solution

Q179. JEE Main 2022 (Online) 27th June Morning Shift

A 220 V, 50 Hz AC source is connected to a 25 V, 5 W lamp and an additional resistance R in series (as shown in figure) to run the lamp at its peak brightness, then the value of R (in ohm) will be _____.

Open solution

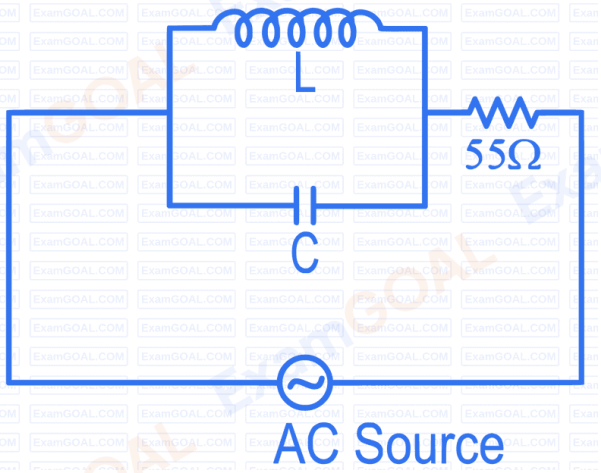


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Q180. JEE Main 2022 (Online) 26th June Morning Shift

A 110 V, 50 Hz, AC source is connected in the circuit (as shown in figure). The current through the resistance 55 Ω, at resonance in the circuit, will be _____ A.

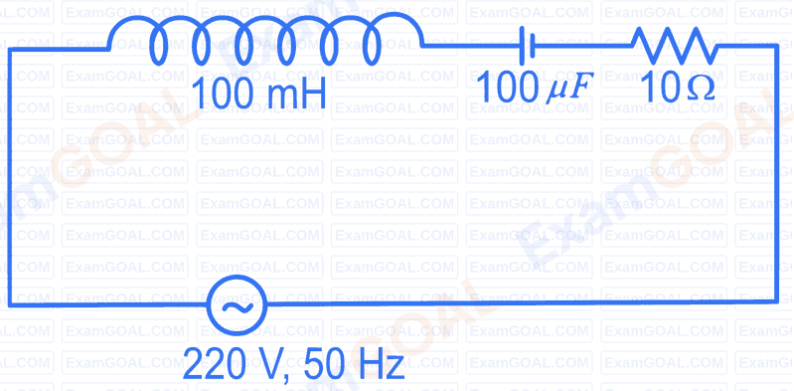
Open solution



Q181. JEE Main 2022 (Online) 25th June Evening Shift

In a series LCR circuit, the inductance, capacitance and resistance are L = 100 mH, C = 100 μF and R = 10 Ω respectively. They are connected to an AC source of voltage 220 V and frequency of 50 Hz. The approximate value of current in the circuit will be _____ A.

Open solution

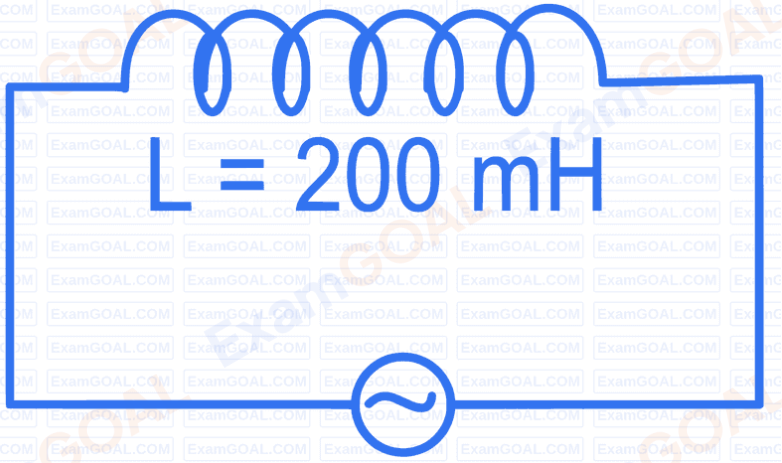


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Q182. JEE Main 2022 (Online) 24th June Morning Shift

As shown in the figure an inductor of inductance 200 mH is connected to an AC source of emf 220 V and frequency 50 Hz. The instantaneous voltage of the source is 0 V when the peak value of current is $\frac{\sqrt{a}}{\pi}$ A. The value of a is _____.

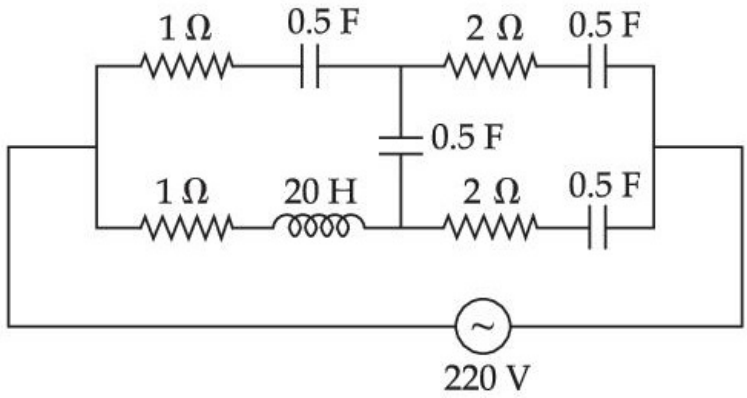
Open solution



Q183. JEE Main 2021 (Online) 31st August Evening Shift

At very high frequencies, the effective impedance of the given circuit will be _____ Ω.

Open solution



Q184. JEE Main 2021 (Online) 27th August Evening Shift

An ac circuit has an inductor and a resistor of resistance R in series, such that X_L = 3R. Now, a capacitor is added in series such that X_C = 2R. The ratio of new power factor with the old power factor of the circuit is $\sqrt{5} : x$. The value of x is _____.

Open solution

Q185. JEE Main 2021 (Online) 27th August Morning Shift

The alternating current is given by $i = \left\{\sqrt{42}\sin\left(\frac{2\pi}{T}t\right) + 10\right\}A$

The r.m.s. value of of this current is ..... A.

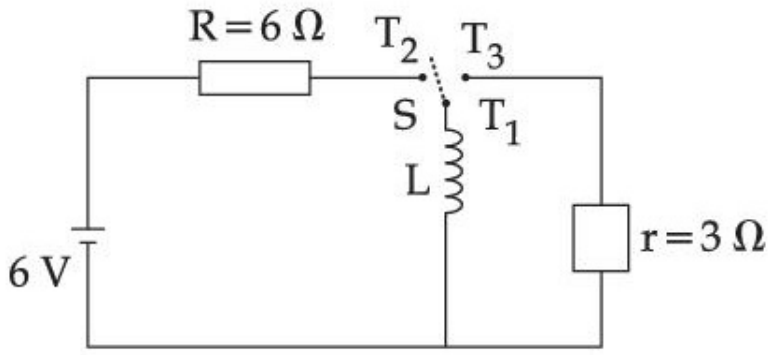
Open solution

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Q186. JEE Main 2021 (Online) 27th July Morning Shift

Open solution

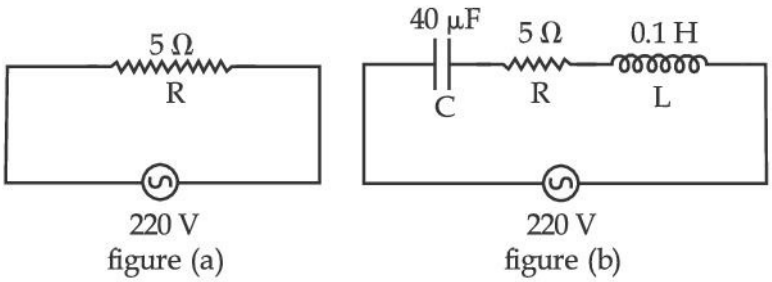
Consider an electrical circuit containing a two way switch 'S'. Initially S is open and then T_1 is connected to T_2. As the current in R = 6Ω attains a maximum value of steady state level, T_1 is disconnected from T_2 and immediately connected to T_3. Potential drop across r = 3Ω resistor immediately after T_1 is connected to T_3 is _____ V. (Round off to the Nearest Integer)



Q187. JEE Main 2021 (Online) 25th July Evening Shift

Open solution

Two circuits are shown in the figure (a) & (b). At a frequency of _____ rad/s the average power dissipated in one cycle will be same in both the circuits.



Q188. JEE Main 2021 (Online) 25th July Morning Shift

Open solution

An inductor of 10 mH is connected to a 20V battery through a resistor of 10 kΩ and a switch. After a long time, when maximum current is set up in the circuit, the current is switched off. The current in the circuit after 1 μs is $\frac{x}{100}$ mA. Then x is equal to _____. (Take e^{-1} = 0.37)

Q189. JEE Main 2021 (Online) 20th July Evening Shift

Open solution

A series LCR circuit of R = 5Ω, L = 20 mH and C = 0.5 μF is connected across an AC supply of 250 V, having variable frequency. The power dissipated at resonance condition is _____ $\times 10^2$ W.

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Q190. JEE Main 2021 (Online) 20th July Morning Shift

In an LCR series circuit, an inductor 30 mH and a resistor 1 Ω are connected to an AC source of angular frequency 300 rad/s. The value of capacitance for which, the current leads the voltage by 45° is $\frac{1}{x} \times 10^{-3}$ F. Then the value of x is _____.

Open solution

Q191. JEE Main 2021 (Online) 16th March Morning Shift

A sinusoidal voltage of peak value 250 V is applied to a series LCR circuit, in which R = 8Ω, L = 24 mH and C = 60 μF. The value of power dissipated at resonant condition is 'x' kW. The value of x to the nearest integer is _____.

Open solution

Q192. JEE Main 2021 (Online) 25th February Morning Shift

A transmitting station releases waves of wavelength 960 m. A capacitor of 2.56 μF is used in the resonant circuit. The self inductance of coil necessary for resonance is _____ $\times 10^{-8}$ H.

Open solution

Q193. JEE Main 2021 (Online) 24th February Evening Shift

A series L-C-R circuit is designed to resonate at an angular frequency $\omega_0 = 10^5$ rad/s. The circuit draws 16W power from 120V source at resonance. The value of resistance 'R' in the circuit is _____ Ω.

Open solution

Q194. JEE Main 2021 (Online) 24th February Morning Shift

A common transistor radio set requires 12 V (D.C.) for its operation. The D.C. source is constructed by using a transformer and a rectifier circuit, which are operated at 220 V (A.C.) on standard domestic A.C. supply. The number of turns of secondary coil are 24, then the number of turns of primary are _____.

Open solution

Q195. JEE Main 2020 (Online) 6th September Evening Slot

In a series LR circuit, power of 400W is dissipated from a source of 250 V, 50 Hz. The power factor of the circuit is 0.8. In order to bring the power factor to unity, a capacitor of value C is added in series to the L and R. Taking the value of C as $\left(\frac{n}{3\pi}\right)$ μF, then value of n is _____.

Open solution

Out of syllabus 9 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q196. JEE Main 2022 (Online) 30th June Morning Shift · MCQ (Single Correct Answer)

In series RLC resonator, if the self inductance and capacitance become double, the new resonant frequency (f_2) and new quality factor (Q_2) will be :

(f_1 = original resonant frequency, Q_1 = original quality factor)

Open solution

(A) $f_2 = \frac{f_1}{2}$ and $Q_2 = Q_1$

(B) $f_2 = f_1$ and $Q_2 = \frac{Q_1}{Q_2}$

(C) $f_2 = 2f_1$ and $Q_2 = Q_1$

(D) $f_2 = f_1$ and $Q_2 = 2Q_1$

Q197. JEE Main 2020 (Online) 6th September Morning Slot · MCQ (Single Correct Answer)

An AC circuit has R= 100 Ω, C = 2 μF and L = 80 mH, connected in series. The quality factor of the circuit is :

Open solution

(A) 20 (B) 2 (C) 0.5 (D) 400

Q198. JEE Main 2018 (Offline) · MCQ (Single Correct Answer)

For an RLC circuit driven with voltage of amplitude vm and frequency $\omega_0 = \frac{1}{\sqrt{LC}}$ the current exhibits resonance. The quality factor, Q is given by :

Open solution

(A) $\frac{CR}{\omega_0}$

(B) $\frac{\omega_0 L}{R}$

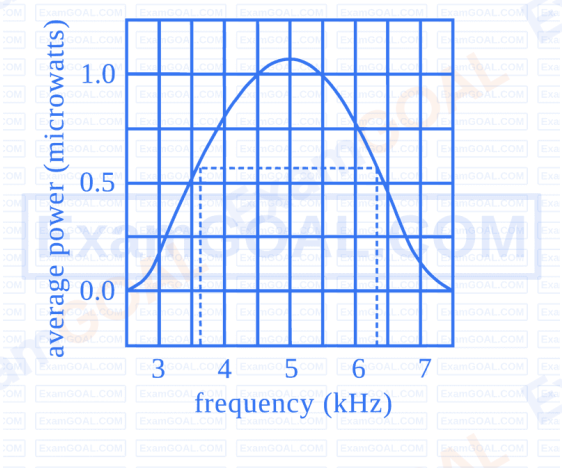
(C) $\frac{\omega_0 R}{L}$

(D) $\frac{R}{(\omega_0 C)}$

Q199. JEE Main 2013 (Online) 23th April Morning Slot · MCQ (Single Correct Answer)

The plot given below is of the average power delivered to an LRC circuit versus frequency. The quality factor of the circuit is :

Open solution



(A) 5.0 (B) 2.0 (C) 2.5 (D) 0.4

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Q200. JEE Main 2013 (Online) 22nd April Morning Slot · MCQ (Single Correct Answer)

In a series L - C - R circuit, $C = 10^{-11}$ Farad, $L = 10^{-5}$ Henry and $R = 100$ Ohm, when a constant D.C. voltage E is applied to the circuit, the capacitor acquires a charge 10^{-9} C. The D.C. source is replaced by a sinusoidal voltage source in which the peak voltage E_0 is equal to the constant D.C. voltage E. At resonance the peak value of the charge acquired by the capacitor will be :

(A) 10^{-15} C (B) 10^{-6} C (C) 10^{-10} C (D) 10^{-8} C

Open solution

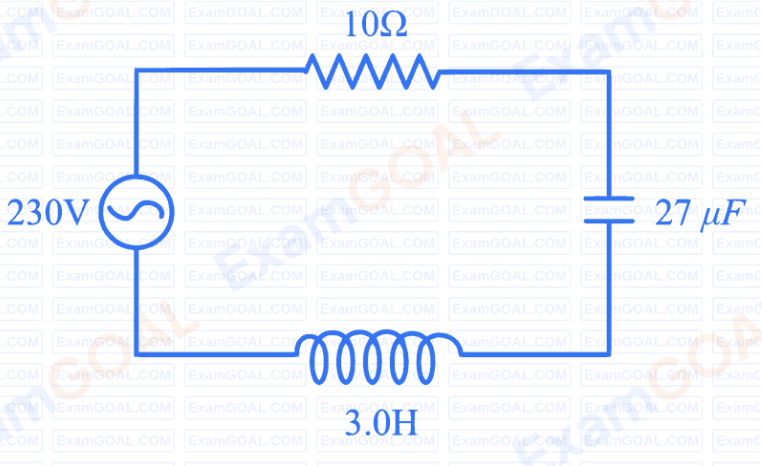
Q201. JEE Main 2023 (Online) 8th April Evening Shift · Numerical

A series combination of resistor of resistance $100\ \Omega$, inductor of inductance $1\ \mathrm{H}$ and capacitor of capacitance $6.25\ \mu\mathrm{F}$ is connected to an ac source. The quality factor of the circuit will be _____

Open solution

Q202. JEE Main 2023 (Online) 24th January Morning Shift · Numerical

In the circuit shown in the figure, the ratio of the quality factor and the band width is _____ s.



Open solution

Q203. JEE Main 2021 (Online) 26th February Morning Shift · Numerical

In a series LCR resonant circuit, the quality factor is measured as 100. If the inductance is increased by two fold and resistance is decreased by two fold, then the quality factor after this change will be _____.

Open solution

Q204. JEE Main 2021 (Online) 24th February Morning Shift · Numerical

A resonance circuit having inductance and resistance 2×10^{-4} H and 6.28Ω respectively oscillates at 10 MHz frequency. The value of quality factor of this resonator is _____. [π = 3.14]

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **A**	2. **A**	3. **C**	4. **B**	5. **B**
6. **C**	7. **C**	8. **D**	9. **A**	10. **D**
11. **B**	12. **B**	13. **B**	14. **A**	15. **C**
16. **B**	17. **D**	18. **C**	19. **A**	20. **B**
21. **C**	22. **B**	23. **C**	24. **D**	25. **B**
26. **A**	27. **C**	28. **C**	29. **C**	30. **A**
31. **A**	32. **C**	33. **C**	34. **D**	35. **D**
36. **C**	37. **D**	38. **C**	39. **B**	40. **A**
41. **C**	42. **A**	43. **B**	44. **D**	45. **C**
46. **C**	47. **C**	48. **D**	49. **C**	50. **B**
51. **B**	52. **B**	53. **C**	54. **B**	55. **C**
56. **D**	57. **D**	58. **A**	59. **A**	60. **B**
61. **C**	62. **A**	63. **C**	64. **C**	65. **B**
66. **A**	67. **D**	68. **A**	69. **A**	70. **C**
71. **A**	72. **D**	73. **C**	74. **A**	75. **A**
76. **A**	77. **D**	78. **C**	79. **C**	80. **D**
81. **D**	82. **A**	83. **C**	84. **B**	85. **C**
86. **B**	87. **A**	88. **D**	89. **A**	90. **A**
91. **C**	92. **B**	93. **D**	94. **C**	95. **C**
96. **C**	97. **C**	98. **A**	99. **A**	100. **B**
101. **C**	102. **B**	103. **C**	104. **C**	105. **B**
106. **B**	107. **B**	108. **D**	109. **D**	110. **A**
111. **B**	112. **C**	113. **B**	114. **C**	115. **C**
116. **D**	117. **D**	118. **C**	119. **C**	120. **C**
121. **D**	122. **A**	123. **D**	124. **B**	125. **C**
126. **C**	127. **C**	128. **D**	129. **D**	130. **A**
131. **B**	132. **A**	133. **D**	134. **A**	135. **C**
136. **A**	137. **B**	138. **B**		

Numerical

139. **200**	140. **100**	141. **314**	142. **50**	143. **1**

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144. **5**	145. **1**	146. **2**	147. **4**	148. **10**
149. **22**	150. **10**	151. **250**	152. **50**	153. **8**
154. **45**	155. **1**	156. **25**	157. **625**	158. **9**
159. **240**	160. **44**	161. **40**	162. **25**	163. **242**
164. **1584**	165. **3872**	166. **8**	167. **100**	168. **3**
169. **100**	170. **10**	171. **250**	172. **3**	173. **44**
174. **500**	175. **5**	176. **0**	177. **15**	178. **100**
179. **975**	180. **0**	181. **22**	182. **242**	183. **2**
184. **1**	185. **11**	186. **3**	187. **500**	188. **74**
189. **125**	190. **3**	191. **4**	192. **10**	193. **900**
194. **440**	195. **400**			

Out of syllabus

196. **A**	197. **B**	198. **B**	199. **B**	200. **D**
201. **4**	202. **10**	203. **283**	204. **2000**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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